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Linking Absolute and Relative Mobility: A Tractable Framework, a Decomposition Method, and an Application to Black–White Mobility Differences

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Abstract

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Keywords: Intergenerational mobility, absolute mobility, relative mobility, income inequality, counterfactual simulation, Bayesian estimation.

1 Introduction

Intergenerational social mobility occupies a central place in public debate and scholarly discussions of social inequality, often invoked as a defining marker of a society's character alongside poverty and income inequality. Metaphors such as the "social elevator" or the "level playing field" frame mobility as a public good—a social trait we want more of. Yet both in public discourse and academic research, "social mobility" carries multiple and sometimes conflicting meanings (Fields and Ok, 1996). Even within the narrower domain of economic mobility, Fields identifies no fewer than ten distinct interpretations (Fields, 2010).

Despite this diversity, discussions of intergenerational mobility tend to coalesce around two perspectives. One asks whether children surpass, match, or fall short of their parents' socioeconomic position—what we call *absolute mobility*. This perspective dominates public debate, where mobility is evaluated in terms of rising or falling living standards across generations. The other asks how strongly children's outcomes depend on those of their parents—what we call *relative mobility*. Relative mobility has been central to academic research as a proxy for equality of opportunity rather than material progress (Breen, 2010).

A growing literature emphasizes that these two dimensions answer fundamentally different questions and need not move together. Societies may exhibit high absolute mobility alongside low relative mobility when broad economic growth raises incomes without reshuffling the hierarchy of advantage. Conversely, they may display rising relative mobility alongside declining absolute mobility when compression occurs through stagnation or decline at the top (Goldthorpe, 2016; Bloome and Opacic, 2024; Bloome, 2025). From this perspective, divergence between absolute and relative mobility is expected, not anomalous. Yet treating them as separate frameworks obscures a basic fact: both are generated by the same underlying process linking parental resources to children's outcomes.

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This paper bridges that divide by embedding absolute and relative mobility within a single, explicitly specified income-generating process. Building on the canonical intergenerational elasticity (IGE) framework, I express individual probabilities of absolute mobility—upward, immobile, and downward—as the probability that a child's *conditional* income distribution exceeds, approximates, or falls short of parental income by a substantively meaningful margin. This conditional distribution is characterized by four components: an income floor (intercept), intergenerational persistence (slope), transmission uncertainty (conditional dispersion), and parental income, which serves as the conditioning baseline. At the aggregate level, societal shares of each mobility type arise from composition-weighted aggregation of these individual probabilities. In this framework, relative mobility corresponds to one important structural feature—the strength of intergenerational persistence—while absolute mobility emerges from the joint interaction of all components.

Previous attempts to connect absolute and relative mobility have typically proceeded from the joint distribution of parent and child incomes. Classical sociological approaches decomposed absolute mobility into marginal distributions and margin-free association structures, treating relative mobility as substantively meaningful and marginal change as nuisance (Hout, 1983; Sobel et al., 1985). More recent economic approaches recover absolute mobility from the joint distribution of income ranks and marginal income growth (Chetty et al., 2017), while Bloome's (2025) recent work derives absolute mobility probabilities from a joint log-normal model of parental and adult incomes. These approaches have clarified why absolute and relative mobility are linked, but they obscure how mobility prospects vary across the parental income distribution, embedding compositional differences mechanically and recovering individual-level heterogeneity only ex post.

The present framework specifies the income-generating process *ex ante* and treats parental income as a conditioning variable rather than a random draw. This shift has several advantages.

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First, it remains aligned with the standard IGE model that has anchored decades of empirical research, ensuring continuity with established practice. Second, it extends that model in a minimal but consequential way by allowing conditional dispersion to vary with parental income, capturing heterogeneity in risk and uncertainty that is empirically salient but often obscured by homoskedastic specifications. Third, because the framework is fully parametric and interpretable, its components map directly onto substantively legible social processes: income floors reflect baseline economic position, intergenerational persistence captures relative immobility, and conditional dispersion reflects uncertainty in income transmission.

This structure makes it possible to derive closed-form expressions linking structural parameters to absolute mobility probabilities and to translate estimated coefficients into transparent statements about mobility regimes. Because these components operate jointly rather than independently—as I formally show—I develop a decomposition of group differences in absolute mobility based on counterfactual parameter swaps. By replacing one component of the income-generating process at a time while holding others fixed, the decomposition isolates the contributions of income floors, intergenerational persistence, and transmission uncertainty. These pointwise effects are then aggregated to the population level while explicitly accounting for compositional differences in parental income distributions. The approach thus avoids the conceptual ambiguities of earlier decompositions by grounding all counterfactuals in a single, coherent generative model.

I validate the framework through a Monte Carlo experiment and then apply it to Black–White differences in absolute mobility in the United States using data from the Panel Study of Income Dynamics. The results show that racial differences are driven by income floors for men and intergenerational persistence for women, while income uncertainty plays a secondary role. Yet the dominant driver of Black–White differences in observed absolute mobility is compositional: the comparatively poorer origins of Black individuals mechanically increase the likelihood of surpassing parental income, while the more advantaged origins of Whites expand the scope for down-

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ward mobility. This application illustrates why integrating absolute and relative mobility is essential. Because absolute mobility is defined relative to parental income, disadvantaged origins mechanically increase the likelihood of upward mobility, while advantaged origins mechanically expand the scope for downward mobility. Whether these mechanical pressures translate into durable advancement or fragile trajectories depends on intergenerational persistence and transmission uncertainty. Seen through this lens, weaker origin–destination associations can coexist with high rates of downward mobility from middle- and high-income origins and with upward mobility from low-income origins (Bloome and Opacic, 2024)—a contemporary interpretation of what has been described as *perverse openness* (Blau and Duncan, 1967; Duncan, 1968; Hout, 1984).

The article proceeds as follows. Section 2 develops the conceptual framework and formal definitions. Section 3 situates the proposed approach relative to existing frameworks. Section 4 derives closed-form expressions showing how income floors, intergenerational persistence, transmission uncertainty, and parental income jointly determine absolute mobility probabilities. Section 5 introduces the counterfactual decomposition and distinguishes structural from compositional effects. Section 6 presents the Bayesian estimation strategy. Section 7 validates the framework through Monte Carlo simulation. Section 8 applies the framework to Black–White differences in absolute mobility using PSID data. The final section discusses implications for theories of mobility, racial inequality, and the interpretation of absolute and relative mobility measures.

2 Conceptual Framework and Definitions

This section develops the conceptual and formal foundations of a unified framework connecting relative and absolute intergenerational mobility. Although these two forms of mobility are typically treated as distinct population-level features, both arise from the same underlying income-

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generating process linking parents' and children's earnings. By explicitly grounding absolute mobility in the structural components of that process—one of which corresponds directly to relative mobility—the framework provides a coherent basis for interpreting group differences in upward, downward, and immobile mobility as the outcome of the complex interaction between intergenerational relative immobility and other structural features of the income transmission process.

At the most general level, intergenerational *relative mobility* captures how closely children's economic standing follows that of their parents. Conceptually, this relationship can be described by the conditional distribution $f(y \mid x)$, where y represents an economic outcome of the child and x that of the parents. The two generations are statistically linked whenever the distribution of Y varies with X. Conversely, if $f(y \mid x) = f(y)$ for all x, parental background and child economic outcomes are independent — a situation that would correspond to perfect relative mobility. Statistically, this conditional relationship can be characterized in multiple ways, depending on how one chooses to represent the distribution of y, summarize the dependence between y and x, and specify the functional form linking them. Existing approaches range from nonparametric descriptions and quantile mappings to regression-based summaries of conditional moments.

In the context of intergenerational income and earnings mobility — arguably the most fruitful domain of mobility research in recent decades — the dominant empirical framework models the transmission process by assuming that $Y \mid X = x$ is approximately log-normal and characterizing the dependence between generations via the conditional expectation function, $\mathbb{E}[Y \mid X = x]$. Specifically, the standard specification assumes a linear relationship in log incomes:

$$y_i \mid x_i \sim \text{Normal}\big(\mu_i = \alpha + \beta x_i,\ \sigma\big),$$

where y_i and x_i denote the natural logarithm of permanent income for children and parents,

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respectively. This log–log model provides a parsimonious summary of intergenerational dependence through the slope coefficient β, commonly referred to as the intergenerational elasticity (IGE). The IGE measures the elasticity of children's expected income with respect to parental income: a 1% increase in parental income is associated with a β% increase in expected children's income. Larger values of β indicate stronger intergenerational persistence (lower relative mobility), whereas smaller values reflect weaker persistence (higher relative mobility).

The other component of the conditional mean—often overlooked in empirical work—is the intercept α. Statistically, α represents the expected income of children whose parents sit at parental income zero. Since zero is rarely a meaningful value for permanent parental income, researchers can re-center (or "de–floor") parental income so that zero corresponds to a sensible lower bound of the distribution. Under this transformation, the intercept becomes substantively meaningful: it captures the expected income of children whose parents are at the bottom of the income distribution. Because β is empirically positive and $\mathbb{E}[y \mid x]$ increases monotonically in x, the intercept defined in this way can be interpreted as an income floor—the baseline level of expected income from which the intergenerational income–generation process begins.

Beyond the conditional mean, the dispersion of outcomes around that mean reflects the degree of uncertainty in the intergenerational transmission process. Standard linear models typically assume homoskedasticity, $\mathbb{V}(Y \mid X = x) = \sigma^2$, but the variance may in practice vary with parental income or other covariates without altering the interpretation of β as a measure of relative mobility. Allowing for heteroskedasticity is especially important for linking relative to absolute mobility because conditional dispersion determines how tightly or loosely children's outcomes cluster around their expected values. Higher dispersion implies greater *unpredictability* in economic attainment given parental position, whereas lower dispersion implies more deterministic trajectories. If, for example, conditional dispersion rises with parental income, children from affluent families experience a wider spread of outcomes — some far exceed expectations while

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others fall short. Conversely, declining dispersion among the well-off would suggest more tightly bound outcomes and stronger predictability at the top, while greater dispersion among disadvantaged families signals higher volatility and more scope for both upward leaps and downward slippage relative to expected paths.

To fix ideas, Figure 1 illustrates intergenerational income patterns for two groups that differ in both the expected level and dispersion of children's incomes across the parental income distribution. Each panel shows the conditional distribution of children's permanent income, y, given parental permanent income, x. The solid black line represents the conditional mean, $\mu = \alpha + \beta x$, summarizing the expected level of children's income at each parental position. The spread of the conditional distribution, $\sigma(x)$, captures uncertainty: wider distributions indicate greater variability in children's outcomes given parental income, while narrower ones reflect more predictable intergenerational trajectories.

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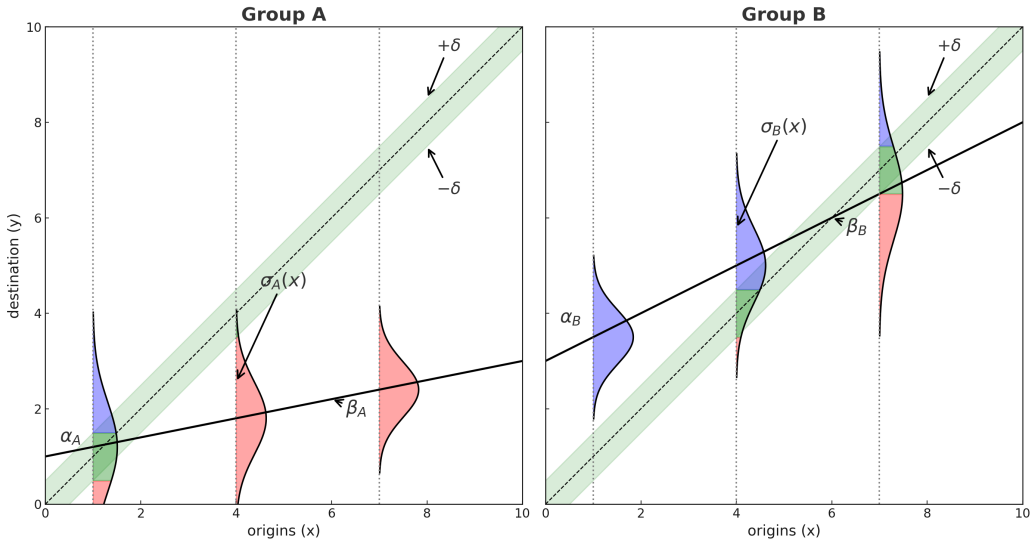

Figure 1: Structural components of intergenerational income transmission: each panel depicts the relationship between parental income (x) and children's expected income (y), given by $\mathbb{E}[y \mid x] = \alpha_g + \beta_g x$. The dashed line marks perfect reproduction ($y = x$); the green band ($\pm\delta$) defines immobility, with blue and red areas representing upward and downward absolute mobility. Normal curves at $x = \{1, 4, 7\}$ illustrate the conditional income distribution $f(y \mid x, g)$, whose spread $\sigma_g(x)$ captures outcome uncertainty. In Group A, a flat slope and declining variance imply weak transmission and compressed outcomes at the top. In Group B, a steeper slope and rising variance reflect stronger persistence and greater volatility, showing how income floors α_g, persistence β_g, and dispersion $\sigma_g(x)$ jointly shape mobility.

Relative mobility in Group A is characterized by a flat conditional mean and a variance that declines with parental income—a regime of weak intergenerational transmission in which parental advantages are only loosely passed on. Higher parental income yields only modest gains for children, and the shrinking variance at the top implies increasingly predictable yet compressed outcomes, reinforcing regression toward the mean. By contrast, Group B exhibits a steeper conditional mean and rising variance across the parental-income distribution, reflecting stronger persistence and a greater imprint of parental resources. The steeper slope signals a stronger capacity to reproduce status, while the widening variance introduces volatility: as parental income rises, outcomes become more dispersed, turning advantage into a domain of both opportunity and risk.

Figure 1 illustrates how absolute mobility differs across two stylized intergenerational income regimes. The solid black line shows the expected income of children, $\mu = \alpha + \beta x$, while the dashed $45°$ line ($y = x$) represents perfect intergenerational reproduction. The green band around the diagonal denotes the immobility threshold ($\pm\delta$), defining a zone of economic stability that avoids the hard knife-edge limit of exact income equality[1]: outcomes above the band ($y - x > \delta$) correspond to upward mobility, and those below ($y - x < -\delta$) to downward mobility. Conditional income distributions $f(y \mid x)$ are shown for selected parental-income levels, with blue, red, and green shading indicating the portions associated with upward, downward, and immobile trajectories, respectively. Together, these visual elements highlight how the structural features of the intergenerational income-generating process jointly determine the likelihood that children rise above, remain near, or fall below their parents' economic position.

In Group A, most individuals experience downward absolute mobility, especially those from higher-income origins. This pattern reflects a modest expected income floor (α_A), weak intergenerational persistence (β_A), and conditional dispersion that narrows as parental income rises (σ_A). Children from disadvantaged backgrounds face limited and uncertain prospects for upward movement, while those from affluent origins encounter a meaningful risk of downward mobility—not because they lack resources, but because when upward drift is weak and outcomes compress at the top, prior advantages are easily lost and difficult to reproduce. By contrast, Group B exhibits substantially higher rates of upward mobility, driven by a higher income floor (α_B) and stronger intergenerational transmission (β_B). However, upward mobility remains origin-dependent. Children from low-income families almost uniformly surpass their parents' income, as they are competing against a lower baseline. At the top of the distribution, however, wider conditional dispersion (σ_B) means children from high-income origins face greater volatility and thus a nontrivial risk of downward movement—advantage brings opportunity but also

[1]The hard knife-edge case of exact income equality is recovered as the limiting case $\delta = 0$: the immobility band collapses to the diagonal, and the probability of immobility is zero by definition.

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exposure to larger swings.

These patterns show that absolute mobility emerges from the interaction between intergenerational persistence (relative immobility) and other structural features of the income-generation process, including expected income floors, uncertainty in the transmission of economic advantage, and the comparison baseline set by parental income. Formally, let groups be indexed by $g \in A, B$; then the probabilities of distinct absolute mobility types—the colored regions in Figure 1—are defined as:

$$\Pr(M^{\uparrow}) = \Pr(y - x > \delta \mid x, g) \qquad \text{(upward mobility)} \qquad (1)$$

$$\Pr(M^{\downarrow}) = \Pr(y - x < -\delta \mid x, g) \qquad \text{(downward mobility)} \qquad (2)$$

$$\Pr(M^{0}) = 1 - [\Pr(M^{\uparrow}) + \Pr(M^{\downarrow})] \qquad \text{(immobility)} \qquad (3)$$

where these probabilities are evaluated under the conditional distribution $f(y \mid x, g; \alpha_g, \beta_g, \sigma_g(x))$, which encapsulates the structural parameters of the intergenerational income-generation process. Together, these parameters determine the chance that a child's income exceeds, approximates, or falls below that of their parents at any given level of x. The next section formalizes how variation in each component shapes absolute mobility patterns within groups and produces systematic differences across them.

2.1 Where is income inequality in this framework?

In the framework developed here, children's income inequality is not introduced as an independent object but instead emerges endogenously from the interaction between the distribution of parental incomes and the parameters governing intergenerational transmission. This feature distinguishes the present approach from methods that treat marginal income distributions as prim-

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itives that can be freely swapped or reweighted independently of the transmission process.

In this setup, the marginal distribution of parental income, $f(x)$, is an exogenous compositional object. Inequality of origins—summarized by $\mathbb{V}(x)$—is therefore not determined by the transmission parameters and can be counterfactually modified through reweighting or distributional swaps (see Section 5, composition). By contrast, inequality in children's incomes is implied by the joint configuration of $f(x)$ and the transmission parameters. Applying the law of total variance yields:

$$\mathbb{V}(y) = \mathbb{E}[\mathbb{V}(y \mid x)] + \mathbb{V}(\mathbb{E}[y \mid x]) = \mathbb{E}[\sigma^2(x)] + \beta^2\,\mathbb{V}(x)$$

This equality clarifies the distinct channels through which inequality in children's outcomes arises. The first term, $\mathbb{E}[\sigma^2(x)]$, captures within-origin dispersion: inequality generated by uncertainty or unpredictability in the income transmission process, averaged over the parental income distribution. The second term, $\beta^2\,\mathbb{V}(x)$, captures between-origin dispersion: inequality mechanically transmitted from unequal starting positions, scaled by the degree of intergenerational persistence.

This formulation highlights why counterfactual analyses in this framework do not proceed by directly imposing alternative levels of inequality in children's incomes. Instead, changes in inequality must be understood as the consequence of changes in the underlying parameters or in the distribution of origins. For example, changing the inequality of parental incomes affects child inequality through both channels, by altering the dispersion of expected outcomes and by reweighting the distribution over which transmission risk is averaged. Changing the persistence parameter β affects inequality exclusively through the between-origin channel, amplifying or dampening the translation of origin inequality into outcomes. Changing the conditional variance function $\sigma^2(x)$ operates through the within-origin channel, modifying the degree of dispersion among

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children with similar parental incomes[2].

In this sense, inequality in the present framework is embedded in the probabilistic structure linking origins to destinations. Unlike approaches that manipulate marginal distributions directly, this framework requires inequality changes to be expressed in terms of concrete transmission mechanisms—persistence, uncertainty, and composition—each of which has distinct implications for both relative and absolute mobility, as well as for observed differences across groups.

3 Comparison with Existing Approaches

This paper contributes to a long-standing effort to clarify the relationship between absolute and relative intergenerational mobility. Although terminology and methods have evolved, the core question remains: how do observed mobility rates relate to the underlying generative processes?

This question originated in sociological research on class mobility (Blau and Duncan, 1967), which distinguished structural mobility (driven by changing occupational distributions) from social fluidity (the net association between origins and destinations). The log-linear modeling tradition formalized this distinction, representing absolute mobility as a function of marginal distributions and a margin-free association structure (Hout, 1983; Sobel et al., 1985). Within this paradigm, analytical attention focused squarely on comparing patterns of fluidity, while marginal distributions were often treated as a nuisance. As Boudon (1973, p. 17) argued, a proper mobility

[2]This decomposition also clarifies the interpretation of a commonly used margin-free measure of intergenerational association, the intergenerational correlation (IGC). In this framework, the Pearson correlation between parental and child log incomes can be written as

$$\rho_{xy} = \beta \frac{\sqrt{\mathbb{V}(x)}}{\sqrt{\mathbb{V}(y)}} = \frac{\beta\sqrt{\mathbb{V}(x)}}{\sqrt{\beta^2\mathbb{V}(x) + \mathbb{E}[\sigma^2(x)]}} = \frac{1}{\sqrt{1+\text{NSR}}}, \qquad \text{where} \quad \text{NSR} \equiv \frac{\mathbb{E}[\sigma^2(x)]}{\beta^2\,\mathbb{V}(x)}.$$

This expression shows that intergenerational correlations depend not only on persistence β, but on the balance between predictable variation transmitted from unequal origins (the "signal," $\beta^2\,\mathbb{V}(x)$) and within-origin uncertainty (the "noise," $\mathbb{E}[\sigma^2(x)]$). A higher noise-to-signal ratio attenuates intergenerational correlations even when persistence is unchanged, motivating the separation of persistence, uncertainty, and composition in the analysis of intergenerational and absolute mobility.

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index should "eliminate" mobility generated by structural change.

This framework spurred foundational work but also sustained debates over how statistical decompositions map onto substantive concepts (Sobel, 1983; Cobalti, 1988; Xie, 1992). A key resolution was the recognition that absolute and relative mobility are not distinct types of mobility but complementary perspectives on the same phenomenon (Cobalti, 1988), with the marginal/association distinction serving primarily as a methodological device rather than a substantive dichotomy.

Later work in economics on income mobility focused predominantly on relative mobility. Its workhorse measure —the intergenerational elasticity (IGE)— quantifies how parental resources translate into children's economic outcomes (Solon, 1992, 1999; Zimmerman, 1992; Black and Devereux, 2010), but remains largely silent on absolute mobility, and on the connection between the two. Only more recently has work by Chetty and coauthors addressed both dimensions simultaneously. By reframing income mobility in terms of the joint distribution of parent and child incomes —expressed as the copula of their percentile ranks combined with marginal income distributions— absolute mobility becomes a derived quantity directly estimable from these two components (2017).

Most recently, Bloome (2025) provides one of the most systematic attempts to formally unpack the individual-level relationship between absolute and relative income mobility. Building on Berman (2022), Bloome models parental and adult incomes as jointly log-normal, deriving absolute mobility probabilities consistent with standard IGE-based models. She decomposes expected absolute mobility into three components: shared growth (average income gains affecting all individuals regardless of parental position), mean reversion (the tendency for individuals from above- or below-mean parental backgrounds to move toward the generational mean), and halted reversion (the extent to which lower relative mobility anchors individuals closer to parental positions). The framework further introduces unexpected absolute mobility, capturing income variation not predicted by parental income. Even under homoskedasticity, Bloome shows, this

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component induces heterogeneity: as relative mobility rises, the variance of unexpected mobility increases and —through nonlinear threshold crossing— generates asymmetric effects on upward mobility probabilities across parental backgrounds. Overall, Bloome demonstrates that observed absolute–relative mobility associations are not mechanical identities but contingent outcomes arising from the joint action of growth, inequality, and intergenerational dependence.

The framework developed here shares important commonalities with Bloome's approach, yet it departs in modeling strategy and interpretive emphasis in substantively consequential ways. Whereas Bloome begins from the joint distribution of parental and adult incomes and recovers individual-level components ex post, the present framework specifies the income-generating process ex ante. Parental income is treated as a conditioning variable rather than as a random draw, and several standard IGE assumptions—largely innocuous for analyses of relative mobility but nontrivial for absolute mobility—are explicitly relaxed. The guiding intuition is straightforward: multidimensional problems are most tractable when decomposed into lower-dimensional components whose roles can be isolated and then recombined. In this respect, Bloome's framework is particularly well suited for statistical decompositions that clarify why and under what conditions absolute and relative mobility are related. By contrast, the present framework is designed to make transparent *for whom*, in terms of social origins, upward, downward, and immobile outcomes occur, and how these outcomes arise from the interaction between relative mobility and other structural features of the income-generating process. These differences in modeling orientation have several important implications.

First, the present framework prioritizes interpretability at the level of mechanisms. In Bloome's framework, quantities such as mean mobility and halted reversion emerge from an algebraic decomposition that clarifies how absolute and relative mobility are linked, but these components do not map cleanly onto distinct processes of intergenerational transmission. By contrast, the present framework parameterizes the conditional distribution of children's income in terms

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of substantively legible components: income floors (intercepts), intergenerational persistence (slopes), and transmission uncertainty (conditional dispersion). As a result, probabilities of upward, downward, and immobile mobility—rather than upward mobility alone—follow directly from interpretable structural parameters.

Second, joint-distribution approaches such as Bloome's and Berman's mechanically embed compositional differences in parental income distributions into estimates of absolute mobility. Under positive income growth, for example, groups with lower average parental income will, all else equal, mechanically exhibit higher rates of upward absolute mobility. Conditioning on parental income makes this compositional channel explicit, avoids conflating composition with structure, and—most importantly—renders visible *for whom*, in terms of social origins, upward, downward, and immobile outcomes actually occur, rather than attributing observed patterns to aggregate mobility alone.

Third, by extending the standard IGE framework to model both the conditional mean and the conditional variance of children's income, the present framework allows transmission risk to vary systematically across the parental income distribution. This heterogeneity is empirically consequential: uncertainty can shape mobility chances differently at the bottom, middle, and top of the distribution, and such patterns are obscured by homoskedastic models.

Fourth, starting from the child income–generating process makes it possible to derive closed-form expressions for the marginal effects of each structural parameter on absolute mobility probabilities and to implement counterfactual simulations that isolate structural mechanisms alongside compositional effects in a transparent and modular way.

In sum, the present approach retains the core insight that absolute and relative mobility are intrinsically connected, but it re-expresses the problem to foreground generative mechanisms, accommodate richer heterogeneity, and align directly with the parameters most commonly estimated in empirical research. By embedding absolute and relative mobility within a unified income-

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generating process that extends the standard IGE framework, it derives their relationship analytically from interpretable structural components and provides tools to isolate specific mechanisms through counterfactual analysis.

4 Drivers of Absolute-Mobility Probabilities

The central goal of this framework is to express absolute mobility probabilities as explicit functions of the structural parameters of the intergenerational income model—transforming absolute mobility from a descriptive summary into a structural outcome. By anchoring both absolute and relative mobility in the same probabilistic system, the framework shows that they emerge from the single income-generation process. Their interdependence becomes analytically transparent rather than conceptually assumed. This shift turns broad empirical questions —such as why some groups experience greater upward mobility or why others face higher risks of downward movement— into tractable, model-based problems. Observed mobility gaps can be decomposed into the distinct contributions of separate structural forces, each operating differently across the parental income distribution.

To formalize this connection, I focus on the structural parameters that govern the conditional distribution of children's income given parental income. The intercept α captures expected income for children from parents at the bottom of the distribution—an income "floor." The slope β reflects the strength of intergenerational transmission (relative immobility). The dispersion σ captures the uncertainty surrounding expected income across parental origins. Parental income x provides the baseline against which children's outcomes are evaluated. I then compute the partial derivatives of the absolute mobility probabilities with respect to α, β, σ, and x, thereby quantifying how marginal shifts in each parameter reshape the likelihood of upward, downward, and immobile mobility.

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With this structure in place, I express the probabilities of absolute upward, downward, and immobile mobility directly as functions of these parameters. Upward mobility is defined as the probability that children's income exceeds their parents' by at least δ; downward mobility is the probability that it falls short by at least δ; and immobility corresponds to remaining within $\pm\delta$ of parental income. To streamline the derivations, I define the standardized distances from the conditional mean to the upward and downward thresholds:

$$z^{\uparrow} = \frac{x + \delta - \mu}{\sigma} = \frac{x + \delta - (\alpha + \beta x)}{\sigma}, \tag{4}$$

$$z^{\downarrow} = \frac{x - \delta - \mu}{\sigma} = \frac{x - \delta - (\alpha + \beta x)}{\sigma}, \tag{5}$$

where $\mu = \alpha + \beta x$ and σ denote the conditional mean and standard deviation of child income, evaluated at parental income x for a given group g (group indices and the dependence of σ on x are omitted for notation clarity). The corresponding mobility probabilities are

$$\Pr\left(M^{\uparrow}\right) = 1 - F(z^{\uparrow}), \tag{6}$$

$$\Pr\left(M^{\downarrow}\right) = F(z^{\downarrow}), \tag{7}$$

$$\Pr\left(M^{0}\right) = F(z^{\uparrow}) - F(z^{\downarrow}), \tag{8}$$

where $F(\cdot)$ and $f(\cdot) = F'(\cdot)$ denote the cumulative distribution function and its associated density.

Having re-expressed the problem in this form, I examine how each structural parameter—α, β, σ, and x—shapes these probabilities by deriving their partial derivatives (see Appendix 10). Because δ defines the width of the immobility band, its effect on mobility probabilities is largely mechan-

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ical: as δ increases, both upward and downward mobility probabilities shrink while immobility rises, not due to any structural change in the income-generating process but simply because fewer children cross more distant thresholds.

The following derivatives show how marginal shifts in each component shape the probability of upward, downward, and immobile mobility, clarifying the distinct role each parameter plays in producing observed mobility patterns:

Income floors. The partial derivatives with respect to the intercept α capture how changes in a group's expected income floor affect absolute mobility probabilities.[3] An increase in α shifts the entire conditional distribution of children's income upward, implying uniform income growth across all levels of parental income. Consequently, the probability of upward mobility rises while the probability of downward mobility falls:

$$\frac{\partial \Pr\left(M^{\uparrow}\right)}{\partial \alpha} = \frac{f(z^{\uparrow})}{\sigma}, \tag{9}$$

$$\frac{\partial \Pr\left(M^{\downarrow}\right)}{\partial \alpha} = -\frac{f(z^{\downarrow})}{\sigma}. \tag{10}$$

Here, $f(z)$ denotes the density of the standardized conditional income distribution evaluated at the relevant mobility threshold. Intuitively, when all child incomes increase by the same amount, more individuals cross the upward-mobility cutoff and fewer cross the downward cutoff. These effects are strongest when the conditional distribution places substantial probability mass near the thresholds—formally, when $f(z)$ is large—and when dispersion is low (σ small). Substantively, this corresponds to a compressed distribution that is steep around the cutoffs, so even modest shifts in mean income generate sizable changes in mobility probabilities. By contrast, when the conditional distribution lies far from the thresholds or is highly dispersed, uniform in-

[3]This interpretation relies on re-centering parental income so that zero represents a meaningful lower bound of its distribution. Under this normalization, the intercept corresponds to the expected income of children whose parents are at the bottom of the income distribution, making α interpretable as an income floor.

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come growth has limited influence on mobility outcomes. Importantly, because the distribution may lie at different distances from the upward and downward cutoffs, a change in overall income need not produce symmetric effects on upward and downward mobility.

The effect on immobility depends on the relative density around the two thresholds, highlighting that changes in economic standing may either increase or decrease immobility depending on whether the upward or downward threshold lies closer to the bulk of the distribution:

$$\frac{\partial \Pr\left(M^0\right)}{\partial \alpha} = \frac{f(z^{\uparrow}) - f(z^{\downarrow})}{\sigma}. \tag{11}$$

Relative immobility: The partial derivatives with respect to the slope parameter β capture how changes in the strength of the association between parental and child income affect probabilities of absolute mobility:

$$\frac{\partial \Pr\left(M^{\uparrow}\right)}{\partial \beta} = \frac{x\, f(z^{\uparrow})}{\sigma}, \tag{12}$$

$$\frac{\partial \Pr\left(M^{\downarrow}\right)}{\partial \beta} = -\frac{x\, f(z^{\downarrow})}{\sigma}. \tag{13}$$

A key insight is that the effect of β operates in proportion to the effect of the intercept α. Whereas α shifts the entire conditional income distribution uniformly, β tilts it: expected child income rises for all families when β increases, but the gains grow with parental income, making the shift substantially larger for higher-x families. This proportionality highlights that β is a positional parameter: the impact of relative immobility on absolute mobility depends directly on where a family begins in the parental income distribution. For children from higher-income families, strengthening persistence raises expected child income sharply, moving the conditional mean

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closer to the upward mobility threshold and farther from the downward one; upward mobility therefore increases and downward mobility declines. For children from lower-income families, increases in β raise expected income only modestly relative to the upward threshold and slightly reduce proximity to the downward one. As a result, stronger persistence also increases upward mobility and reduces downward mobility among low-income families, but the magnitude of these changes is comparatively small.

The effect of persistence on immobility follows directly from the difference in probability density at the downward and upward thresholds:

$$\frac{\partial \Pr(M^0)}{\partial \beta} = \frac{x\,[f(z^{\downarrow}) - f(z^{\uparrow})]}{\sigma}.$$

Because the expression depends on the relative density at the two boundaries, this derivative is not sign-definite. Its absolute magnitude is greatest when the conditional mean is close to the immobility band, where density is high and small tilts in the conditional mean can shift outcomes across a boundary. Farther from either threshold, $f(z)$ becomes small and the influence of β on immobility diminishes accordingly.

Uncertainty in transmission.

The partial derivatives with respect to the dispersion parameter σ describe how changes in the spread of child income conditional on parental income (transmission uncertainty) shape the probabilities of absolute mobility:

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$$\frac{\partial \Pr\left(M^{\uparrow}\right)}{\partial \sigma} = \frac{z^{\uparrow} f(z^{\uparrow})}{\sigma}, \tag{14}$$

$$\frac{\partial \Pr\left(M^{\downarrow}\right)}{\partial \sigma} = -\frac{z^{\downarrow} f(z^{\downarrow})}{\sigma}, \tag{15}$$

$$\frac{\partial \Pr\left(M^{0}\right)}{\partial \sigma} = \frac{-z^{\uparrow} f(z^{\uparrow}) + z^{\downarrow} f(z^{\downarrow})}{\sigma}. \tag{16}$$

Since both $f(z)$ and σ are strictly positive, the sign of the effect of transmission uncertainty is governed by the standardized distances to the mobility thresholds. The consequences of increased dispersion therefore depend on where the conditional mean lies relative to the upward and downward boundaries. Because $z^{\uparrow}$ and $z^{\downarrow}$ encode these distances, both the sign and magnitude of the derivatives exhibit strong nonlinearities that reflect proximity to each threshold, with magnitudes additionally shaped by the density $f(z)$.

For example, if the expected outcome is far below the upward threshold ($z^{\uparrow}$ is large and positive) but just below the downward threshold ($z^{\downarrow}$ is small and positive), additional dispersion increases upward mobility while reducing downward mobility, with upward effects dominating. In this region, increased uncertainty reduces extreme downside risk while permitting rare but consequential large upward realizations, producing a lottery–ticket effect driven by tail reallocation. By contrast, if the expected outcome lies near or above the upward threshold ($z^{\uparrow}$ is small and negative and $z^{\downarrow}$ is also negative but larger in magnitude), the dominant effect of increased dispersion is a rise in downward mobility accompanied by a comparatively smaller decline in upward mobility. Because the expected outcome already approaches or exceeds the upward boundary, additional dispersion has limited capacity to generate further upward moves while substantially increasing the risk of falling below the downward threshold.

Overall, transmission uncertainty operates positionally: it asymmetrically expands opportunity when expected outcomes lie well below the upward threshold, generates volatility when out-

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comes fall between thresholds, and asymmetrically heightens downside risk as expected outcomes approach or exceed the upward boundary. This positional effect corresponds closely to what Bloome terms unexpected absolute mobility (Bloome, 2025) but here it is explicitly parameterized as a feature of the income-generating process rather than recovered from the joint distribution of parents and children income.

Parental income. Finally, the partial derivatives with respect to parental income x describe how the probabilities of absolute mobility vary across the parental income ladder. Even for a fixed level of intergenerational persistence β, mobility trajectories differ systematically for children of low-, middle-, and high-income parents. These derivatives are:

$$\frac{\partial \Pr\left(M^{\uparrow}\right)}{\partial x} = \frac{\beta - 1}{\sigma} f(z^{\uparrow}), \tag{17}$$

$$\frac{\partial \Pr\left(M^{\downarrow}\right)}{\partial x} = \frac{1 - \beta}{\sigma} f(z^{\downarrow}), \tag{18}$$

$$\frac{\partial \Pr\left(M^{0}\right)}{\partial x} = \frac{1 - \beta}{\sigma} \left[f(z^{\uparrow}) - f(z^{\downarrow})\right]. \tag{19}$$

These expressions show how changes in parental income shift the conditional distribution of children's outcomes relative to fixed mobility thresholds. Because empirical estimates of β typically lie between zero and one, $\partial \Pr\left(M^{\uparrow}\right)/\partial x$ is negative: as parental income rises, the upward threshold moves away from the conditional mean faster than the mean itself rises, reducing the probability of upward mobility. Conversely, $\partial \Pr\left(M^{\downarrow}\right)/\partial x$ is positive: children from higher-income families face a greater risk of downward mobility, since imperfect income reproduction makes earning less than one's parents progressively more likely as parental income increases.

In short, when parental socioeconomic status is only imperfectly reproduced, individuals from lower-income origins face greater scope to surpass their parents, while those from higher-income origins face greater scope to fall below them. The derivative of immobility, $\partial \Pr\left(M^{0}\right)/\partial x$, re-

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flects the balance of these opposing forces and therefore varies non-monotonically across the parental income distribution. Put differently, surpassing low-income parents is mechanically easier than surpassing high-income parents, while falling below high-income parents is mechanically more likely than falling below low-income parents.

Taken together, these derivatives provide a transparent analytical bridge between the structural parameters of the intergenerational income–generating process and resulting patterns of absolute mobility. What emerges is an intrinsically intertwined process: the contribution of any single mechanism depends on the levels of the others. In this sense, the present framework converges with Bloome's (Bloome, 2025)—via a different route—on a key implication: the relationship between absolute and relative mobility is contingent rather than universal.

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5 Counterfactual Decomposition via Parameter Swaps

The analytical derivatives in Section 4 show that the effects of the key structural parameters on absolute mobility probabilities are fundamentally intertwined. Although these derivatives can be evaluated at sample means or other fixed values to summarize marginal effects, such evaluations capture only local sensitivities and do not directly reveal how changes in structural parameters reshape aggregate mobility patterns within and across populations.

To identify the contribution of each structural component in a principled way, I propose a counterfactual simulation approach based on parameter swaps. This method evaluates absolute mobility probabilities under hypothetical scenarios in which a *single* structural parameter is replaced with a counterfactual value —taken from another group, another time period, or a theoretically relevant benchmark— while all other parameters remain fixed at their group-specific values. The goal is to isolate the marginal contribution of each structural parameter to observed mobility differences, net of the others. Applied to comparisons across groups, this framework answers ques-

tions such as: "How much upward mobility would Group A experience if, holding everything else constant, it had the same intergenerational persistence as Group B?"

The simulation proceeds as follows. Let $\delta \geq 0$ denote the tolerance threshold that defines absolute mobility. A child is classified as immobile if $|y - x| < \delta$; otherwise the child experiences upward or downward mobility. For each group g and parental income level x, let the conditional mean of child income be $\mu_g(x) = \alpha_g + \beta_g x$, and let the conditional standard deviation be a generic dispersion function $\sigma_g(x)$. Let F denote the cumulative distribution function of child income. Then, as derived earlier, the observed mobility probabilities for group g at parental income level x are:

$$\overset{(g)}{\Pr}(M^{\uparrow} \mid x) = 1 - F\left(\frac{x + \delta - \mu_g(x)}{\sigma_g(x)}\right), \tag{20}$$

$$\overset{(g)}{\Pr}(M^{\downarrow} \mid x) = F\left(\frac{x - \delta - \mu_g(x)}{\sigma_g(x)}\right), \tag{21}$$

$$\overset{(g)}{\Pr}(M^{0} \mid x) = F\left(\frac{x + \delta - \mu_g(x)}{\sigma_g(x)}\right) - F\left(\frac{x - \delta - \mu_g(x)}{\sigma_g(x)}\right). \tag{22}$$

Next, I simulate counterfactual mobility probabilities $\widetilde{\Pr}^{(g,\theta)}(M^k \mid x)$, where $\theta \in \{\alpha, \beta, \sigma\}$ denotes the parameter being swapped and $k \in \{\uparrow, \downarrow, 0\}$ denotes the mobility type. The counterfactual replaces parameter θ_g for group g with the corresponding value $\theta_{\bar{g}}$ from a comparison group $\bar{g} \neq g$, while all other parameters remain fixed at their original group-g values. Formally:

- α**-swap:** use $\alpha_{\bar{g}}$ in place of α_g, holding β_g and $\sigma_g(x)$ fixed;
- β**-swap:** use $\beta_{\bar{g}}$ in place of β_g, holding α_g and $\sigma_g(x)$ fixed;
- σ**-swap:** use $\sigma_{\bar{g}}(x)$ in place of $\sigma_g(x)$, holding α_g and β_g fixed.

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Having computed the factual and counterfactual probabilities of mobility type k at income level x for group g, denoted $\Pr^{(g)}(M^k \mid x)$ and $\widetilde{\Pr}^{(g,\theta)}(M^k \mid x)$ respectively, the pointwise effect of swapping parameter θ at parental income level x is defined as

$$\Delta_k^{(g,\theta)}(x) = \widetilde{\Pr}^{(g,\theta)}(M^k \mid x) - \overset{(g)}{\Pr}(M^k \mid x). \tag{23}$$

In other words, $\Delta_k^{(g,\theta)}(x)$ measures the change in absolute mobility probabilities produced by replacing θ_g with $\theta_{\bar{g}}$ at parental income level x.

This framework clarifies the role of each structural feature —income floors (α), persistence (β), and transmission uncertainty (σ)— while keeping all other group characteristics and parental income constant. Yet these pointwise comparisons do not by themselves indicate how much each parameter contributes to overall mobility differences across groups. Population-level mobility reflects both how mobility probabilities behave at each parental income level and how many families lie at each point in the parental income distribution. To evaluate how each structural factor contributes to population-wide patterns of absolute mobility, I aggregate these pointwise, income-conditional effects into two complementary summary measures.

Aggregation measures how much, on average, mobility in group g would change if it retained its own parental income distribution but adopted parameter θ from group $\bar{g}$. It averages the pointwise effects $\Delta_k^{(g,\theta)}(x)$ over the empirical income distribution of group g:

$$\overline{\Delta}_k^{(g,\theta)} = \int \Delta_k^{(g,\theta)}(x)\, f^{(g)}(x)\, dx, \tag{24}$$

where $f^{(g)}(x)$ denotes the density of parental income for group g.

Composition isolates the contribution of the parental income distribution itself. It measures how mobility in group g would change if its distribution of parental income were replaced by that of

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group $\bar{g}$, while holding all of g's structural parameters fixed:

$$\overline{\Delta}_k^{(g,x)} = \int p_k^{(g)}(x) \left[f^{(\bar{g})}(x) - f^{(g)}(x) \right] dx, \tag{25}$$

where $p_k^{(g)}(x)$ denotes group g's probability of mobility type k at income level x. This quantity captures the distributional component of mobility differences—the portion attributable solely to differences in where families are located in the parental income distribution, not to how income is transmitted across generations. Together, aggregation and composition translate localized, x-specific effects into population-level statements, separating group differences in absolute mobility into structural and distributional components.

A critical caveat must be stressed: these counterfactuals are descriptive rather than causal. They treat each parameter as if it could vary independently, abstracting away from dependencies among parameters, spillovers, or equilibrium forces that may operate in real-world settings. One can imagine alternative counterfactual regimes that relax this modularity in different ways. At one end of the spectrum are approaches that endogenize the relationship between average child income and intergenerational persistence, motivated by the theoretical claim that more mobile societies may achieve higher aggregate welfare by reallocating talent more efficiently and unlocking otherwise underutilized productive potential(Boudon, 1974; Corak, 2013). In such frameworks, changes in relative mobility mechanically translate into changes in mean income—an empirically estimable relationship (see Bloome, 2025 for one such approach)—thereby embedding absolute mobility within a broader equilibrium process. At the opposite extreme are counterfactuals that impose the reverse restriction, jointly adjusting the intercept and slope so that the mean of the child-income distribution is held fixed—an iso–mean transformation. Iso–mean adjustments alter the interpretation of the structural parameters by forcing them to compensate for one another, mechanically undoing the very shifts in economic position and intergenerational persistence that

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the decomposition seeks to isolate.[4]

The approach adopted here lies between these extremes. I treat the intercept, slope, and variance components as freely varying structural features, favoring analytical transparency and modularity over equilibrium completeness. This choice allows each mechanism to be evaluated in isolation, clarifying how income floors, intergenerational persistence, and transmission uncertainty independently shape absolute mobility. While this framework abstracts from endogenous feedbacks between mobility and aggregate income, it can be straightforwardly extended to incorporate such mechanisms when the substantive goal shifts from decomposition to equilibrium analysis.

Finally, it is important to note that this parameter-swap decomposition is formulated in general terms and does not rely on any particular outcome distribution or functional form for the conditional mean or variance. While the derivations here use the standard IGE specification, any parametric model with an intercept and a parameter governing intergenerational transmission can be used. By expressing absolute mobility probabilities as functions of a generic cumulative distribution F, the method can accommodate arbitrary choices for both mean and dispersion components, making it broadly applicable across heterogeneous subpopulations, time periods, or institutional settings. Moreover, the same logic extends naturally to semiparametric and nonparametric approaches: one may swap empirical conditional means and dispersions, $\mu_g(x)$ and $\sigma_g(x)$, for their counterfactual counterparts without imposing a parametric structure —though this would come at the cost of relinquishing interpretable structural parameters.

[4] An iso–mean counterfactual adjusts the intercept to offset a counterfactual change in the slope so that a group's expected log income is preserved. Let $\bar{x}_g$ denote the mean parental log income for group g. With $\mu_g(x) = \alpha_g + \beta_g x$ and $\bar{\mu}_g = \alpha_g + \beta_g \bar{x}_g$, imposing a counterfactual slope $\beta_g^\star$ requires setting $\alpha_g^\star = \bar{\mu}_g - \beta_g^\star \bar{x}_g = \alpha_g + (\beta_g - \beta_g^\star)\bar{x}_g$, which ensures $\mathbb{E}_g[\alpha_g^\star + \beta_g^\star x] = \bar{\mu}_g$.

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6 Model and Estimation

To characterize the conditional distribution of child income given parental income, I begin with the standard intergenerational elasticity (IGE) framework but relax the common assumption of constant conditional dispersion, allowing income uncertainty to vary systematically with parental income. Formally, I assume

$$y \mid g, x \sim \text{Normal}\big(\mu_g(x), \sigma_g(x)\big), \tag{26}$$

where y and x denote the de-floored natural logarithm of permanent income for children and parents, respectively, and g indexes group membership. Throughout, g_i is a binary indicator taking value 1 for members of the focal group and 0 otherwise. In this formulation, the conditional mean governs expected intergenerational transmission, while the conditional standard deviation captures the uncertainty around that expectation. The conditional mean of child income is modeled as a group-specific linear function of parental income:

$$\mu_{i[g]} = \alpha_g + \beta_g x_i, \qquad \text{with} \qquad \alpha_g = \alpha_0 + \alpha_1 g_i, \quad \beta_g = \beta_0 + \beta_1 g_i \tag{27}$$

Here, α_g denotes the intercept of the conditional mean function for group g, which is interpretable as an income floor—the expected income of children whose parents were at the bottom of the income distribution—provided that $\beta_g > 0$, so that child income increases with parental income. The slope coefficient β_g captures the intergenerational income elasticity for group g, quantifying the strength of association between parental resources and children's expected outcomes. Higher values of β_g indicate stronger persistence and, consequently, lower relative mobility. The conditional standard deviation of child income is modeled flexibly using a lognormal specification to ensure positivity and to allow for heterogeneity in income risk:

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$$\sigma_{i[g]} \mid x_i \sim \text{LogNormal}\big(\gamma_g + \lambda_g x_i,\ \sigma_{\text{disp}}\big)\,, \qquad \text{with} \qquad \gamma_g = \gamma_0 + \gamma_1 g_i, \quad \lambda_g = \lambda_0 + \lambda_1 g_i \tag{28}$$

In this specification, γ_g reflects baseline income uncertainty for children whose parents were at the bottom of the income distribution within group g, while λ_g governs how this uncertainty evolves along the parental income spectrum. The parameter σ_{disp} captures residual heterogeneity in income dispersion that is not explained by parental income or group membership, reflecting idiosyncratic variation in exposure to income risk. Together, these components allow groups to differ not only in their average expected outcomes, but also in the predictability and dispersion of intergenerational income transmission.

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Bayesian Estimation

Model parameters are estimated via Bayesian inference using Hamiltonian Monte Carlo with the No-U-Turn Sampler (NUTS), implemented in Stan. The model adopts a hierarchical structure in which group-specific parameters for both the conditional mean and the log-dispersion function are drawn from population-level distributions with shared hyperparameters.

To balance flexibility, regularization, and computational stability, I assign moderately informative priors to all population-level hyperparameters. For the intercept of the conditional mean and for the log-dispersion coefficients, I specify $\alpha_{\text{mean}} \sim \text{Normal}(0, 0.7)$, $\gamma_{\text{mean}} \sim \text{Normal}(0, 0.7)$, and $\lambda_{\text{mean}} \sim \text{Normal}(0, 0.2)$. For the slope parameter governing intergenerational persistence, I incorporate substantive prior knowledge: $\beta_{\text{mean}} \sim \text{Normal}(0.5, 0.2)$. This prior reflects what Mitnik et al. (2019) refer to as the "dominant hypothesis"—namely, that intergenerational income elasticities in the United States typically lie between 0.3 and 0.6—while remaining weakly informative enough to permit broader variation.

Between-group heterogeneity is governed by population-level standard deviations, which are assigned half-normal priors: $\alpha_{\text{sd}},\ \beta_{\text{sd}},\ \gamma_{\text{sd}},\ \lambda_{\text{sd}} \sim \text{Normal}^{+}(0, \tau)$, with $\tau = \{0.5,\ 0.2,\ 0.3,\ 0.1\}$, respectively. In addition, residual heterogeneity in conditional dispersion is governed by a scale parameter with prior $\sigma_{\text{disp}} \sim \text{Normal}^{+}(0, 0.15)$, which controls the magnitude of idiosyncratic shocks in $\log \sigma_{ig}$ beyond those explained by parental income and group membership.

The model employs a non-centered parameterization, constructing group-specific parameters as scaled deviations from population-level means. This approach improves sampling efficiency and posterior geometry. Full model specification and prior details are provided in the Appendix.

Posterior mobility probabilities

Posterior draws from the fitted model are used to compute absolute mobility probabilities for each observation. Upward and downward mobility are defined using a symmetric log-threshold of $\delta = \log(1.15)$, corresponding to a $\pm 15\%$ band around (unlogged) parental income. For each posterior draw, I compute the predicted probabilities of upward, downward, and immobile mobility as:

$$\Pr(M_i^{\uparrow}) = 1 - \Phi\left(\frac{x_i + \delta - \mu_i}{\sigma_i}\right), \tag{29}$$

$$\Pr(M_i^{\downarrow}) = \Phi\left(\frac{x_i - \delta - \mu_i}{\sigma_i}\right), \tag{30}$$

$$\Pr(M_i^{0}) = \Phi\left(\frac{x_i + \delta - \mu_i}{\sigma_i}\right) - \Phi\left(\frac{x_i - \delta - \mu_i}{\sigma_i}\right), \tag{31}$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function, and $\mu_i = \alpha_g + \beta_g x_i$ and $\sigma_i = \exp(\gamma_g + \lambda_g x_i)$ represent the conditional mean and standard deviation of child income for individual i in group g. Because these probabilities are computed separately for each

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posterior sample, the full distribution of mobility probabilities is obtained for every individual. This structure enables the direct computation of posterior credible intervals for absolute upward, downward, and immobile mobility —quantifying statistical uncertainty around both individual- and group-level estimates.

In addition to computing probabilities under the observed group-specific parameters, I simulate counterfactual mobility scenarios by replacing one or more structural parameters (e.g., α_g, β_g, σ_i) with values drawn from a reference group, as described in Section 5. This procedure isolates the distinct contribution of specific mechanisms —economic starting position, intergenerational transmission strength, and income volatility— to observed group differences in absolute mobility outcomes.

In the next section, I apply this framework to Monte Carlo simulations to evaluate model performance under known data-generating processes, and then to empirical data from the PSID to analyze Black–White differences in absolute income mobility.

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7 A Monte Carlo Experiment

To assess the ability of the proposed framework to recover the structural drivers of intergenerational mobility, I conduct a Monte Carlo experiment based on a fully specified data-generating process (DGP). The simulation features two groups—*Group A* and *Group B*—that differ in their income floors, intergenerational persistence, and the level and evolution of conditional income dispersion. Each group consists of $N = 5{,}000$ individuals.

Parental income x_i^* is generated on the level scale as $x_i^* = \exp(u_i), \qquad u_i \sim \text{Uniform}(0, 12)$. By construction, $x_i = \log x_i^* \geq 0$, so the lower bound of parental log income is normalized to zero. This normalization implies that the intercepts of the conditional income functions are directly interpretable as expected log income for children originating from the bottom of the

parental income distribution, without requiring any additional re-centering or re-flooring. Child income is generated from a Gaussian income-generating process defined on the log scale (so that y denotes log income throughout). Conditional on parental log income x_i and group membership g, children's permanent log income follows

$$y_i \mid x_i, g \sim \text{Normal}\big(\mu_{ig},\ \sigma_{ig}^2\big)\ , \tag{32}$$

$$\mu_{ig} = \alpha_g + \beta_g x_i, \tag{33}$$

$$\log \sigma_{ig} = \gamma_g + \lambda_g x_i + u_{ig}, \qquad u_{ig} \sim \text{Normal}\big(0,\ \sigma_{\text{disp}}\big)\ . \tag{34}$$

This specification allows both the conditional mean and dispersion of children's log income to vary systematically with parental income while remaining fully Gaussian.

The DGP is designed to generate contrasting mobility regimes. Group A exhibits a lower income floor but stronger intergenerational persistence, with parameters $(\alpha_A, \beta_A) = (1.5, 0.70)$. Group B, by contrast, has a higher income floor but displays weaker persistence, with $(\alpha_B, \beta_B) = (2.5, 0.35)$. At the same time, income transmission in Group A is more uncertain than in Group B, with baseline dispersion declining in parental income for Group A and rising for Group B, governed by $(\gamma_A, \lambda_A) = (0.7, -0.1)$ and $(\gamma_B, \lambda_B) = (-0.5, 0.1)$, respectively. The DGP is intentionally stylized to induce pronounced heterogeneity in both expected outcomes and income risk. The model is estimated via Bayesian inference using Stan. The specification incorporates group-specific linear predictors for both the conditional mean and the conditional log standard deviation of income. For each group, four structural parameters are estimated: the intercept (α_g) and slope (β_g) governing the conditional mean, and the intercept (γ_g) and slope (λ_g) governing the conditional dispersion.

Table 1 reports posterior summaries from the Monte Carlo experiment. Posterior means are

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tightly concentrated around the true data-generating values, with narrow 95% credible intervals, indicating accurate and efficient recovery of the structural parameters governing the conditional mean and dispersion[5]

Table 1: Posterior means and 95% credible intervals. True DGP values in parentheses.

	Parameter	Mean	95% CI	
Panel μ. Conditional mean parameters				
Group A	α_A	1.46	(1.38, 1.54)	[1.50]
	β_A	0.71	(0.70, 0.71)	[0.70]
Group B	α_B	2.47	(2.43, 2.51)	[2.50]
	β_B	0.36	(0.35, 0.37)	[0.35]
Panel σ. Conditional dispersion parameters				
Group A	γ_A	0.70	(0.66, 0.74)	[0.70]
	λ_A	-0.10	(-0.11, -0.09)	[-0.1]
Group B	γ_B	-0.48	(-0.53, -0.44)	[-0.50]
	λ_B	0.10	(0.09, 0.11)	[0.10]

Based on posterior draws from the fitted model, I compute the probabilities of upward, downward, and immobile mobility at each parental income level. Immobility is defined using a symmetric log-threshold of $\delta = \log(1.15)$, which corresponds to a $\pm 15\%$ band in unlogged income levels.

To evaluate the contribution of specific mechanisms, I implement a series of parameter-swap counterfactuals that replace one structural component at a time while holding the others fixed. These include (i) an intercept swap (α), (ii) a slope swap (β), and (iii) a variance swap involving both γ and λ. Figure 2 summarizes these results. Panel A displays posterior mean probabilities

[5]Sampling was conducted using four parallel chains with 3,000 iterations each, including 1,000 warmup iterations. Convergence diagnostics indicate stable estimation: all $\hat{R}$ values are approximately 1.00, effective sample sizes exceed 4,000 per chain, and posterior draws exhibit no evidence of autocorrelation or divergent transitions. Estimation employed non-centered parameterizations and weakly informative priors to facilitate efficient exploration of the posterior. The target acceptance rate was set to 0.95 and the maximum tree depth to 12, eliminating divergent transitions and yielding stable energy diagnostics (E-BFMI > 0.3 across all chains).

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of upward, downward, and immobile mobility by group and parental income. Panel B presents pointwise differences between counterfactual and observed mobility probabilities (counterfactual minus observed), with 95% credible intervals, for each group and swap scenario.

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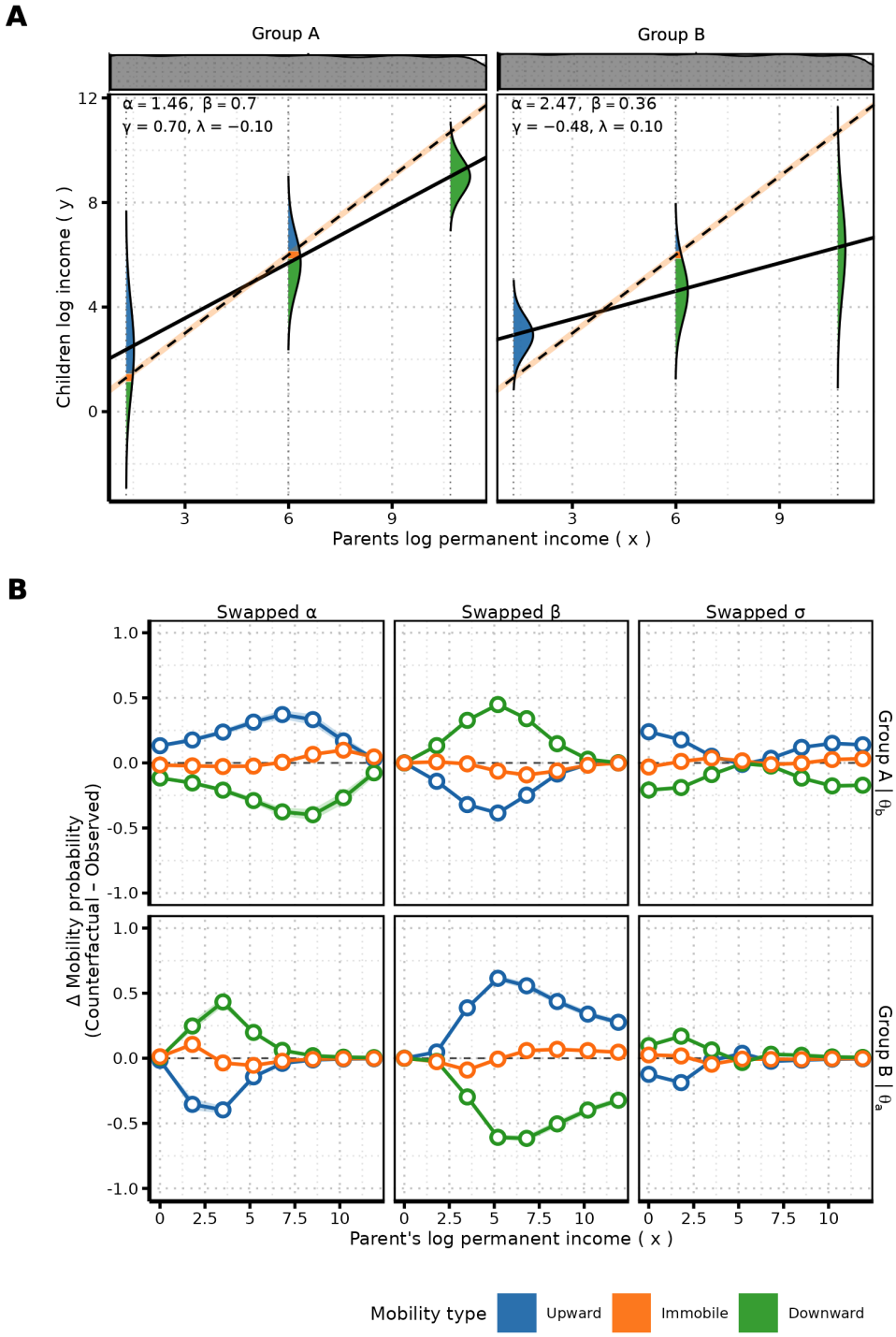

Figure 2: Panel **A** shows the posterior mean probabilities of observed upward, downward, and immobile mobility, by group and parental income. Panel **B** shows pointwise differences in mobility probabilities (counterfactual − observed) with 95% credible intervals, by group and swap.

Panel A reveals that group differences in absolute mobility reflect the joint effects of income floors, intergenerational transmission strength, and income volatility. In Group A, a low income floor and strong intergenerational dependence produce moderate upward mobility from disadvantaged origins, but also introduce substantial downward risk due to elevated income uncertainty. As parental income increases, expected child income rises steeply but not enough to sustain high advantage, leading to a high probability of downward mobility at the top of the distribution. Group B, in contrast, combines a higher income floor with weaker intergenerational transmission and increasing variance. Children from low-income families almost universally experience upward mobility due to elevated expected income and low downside risk. However, because expected income rises slowly with parental income, children from middle- and high-income origins face a high likelihood of downward mobility, amplified by increasing dispersion.

Intercept swaps shift the entire conditional distribution upward or downward. Replacing α_A with the higher α_B in Group A raises expected child income uniformly across all x, increasing upward mobility and reducing downward mobility—except among children from high-income origins, for whom expected income remains below parental levels (top-left panel, Fig. 2). The reverse swap in Group B, substituting α_B with α_A, lowers expected incomes and reduces upward mobility primarily among low- and middle-income families, with muted effects at the top where downward mobility is already saturated (bottom-left panel).

Slope swaps exhibit more heterogeneity across the income distribution. For Group A, adopting Group B's flatter β reduces expected income growth with x, raising downward mobility among middle-income children while modestly decreasing immobility. Low- and high-income origins are largely unaffected: children from disadvantaged origins remain upwardly mobile, while those at the top remain downwardly mobile (top-center panel). In Group B, the introduction of Group A's steeper β increases upward mobility and decreases downward mobility among middle-income families, with minimal change at the extremes (bottom-center panel).

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Variance swaps reallocate probability mass across the tails of the conditional distribution. For Group A, replacing its declining variance function with Group B's increasing $\sigma(x)$ raises upward mobility and reduces downward mobility across most of the income spectrum (top-right panel). This shift is driven by greater dispersion at middle and high parental incomes, which weakens the near-deterministic erosion of advantage observed under low variance. Outcomes are no longer tightly compressed below parental income: while most children from middle- and high-income families still experience downward mobility, a non-negligible fraction escape this otherwise mechanical decline and surpass the upward mobility threshold. In Group B, adopting Group A's flatter variance profile slightly reduces upward mobility and increases downward mobility, primarily among children from disadvantaged origins (bottom-right panel). The narrowing of the conditional distribution in this region constrains upward leaps and introduces greater fragility.

Across all scenarios, changes in immobility are small and highly localized. When the conditional mean moves closer to a mobility threshold, immobility may increase due to higher density near the boundary. When probability mass shifts away from the $\pm\delta$ band, immobility tends to decline. Overall, however, the effects on immobility are modest in both magnitude and scope.

While the panels above report pointwise mobility differences across parental income levels x, these effects must be aggregated to evaluate their population-level consequences. Since structural effects vary with x, and groups often differ in their distribution of parental income, I compute parental-income-weighted averages of the pointwise counterfactual differences using each group's empirical income distribution as weights.

To isolate the contribution of origin composition, I also simulate a counterfactual that holds each group's structural parameters fixed while swapping the parental income distribution. Because the data-generating process assigns the same $x \sim \exp(\mathrm{Unif}(0, 12))$ in both groups, this composition-only swap is expected to produce no difference by design. Figure 3 presents income-weighted posterior mean differences (bars) along with 95% credible intervals (whiskers), decom-

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posed by parameter and group.

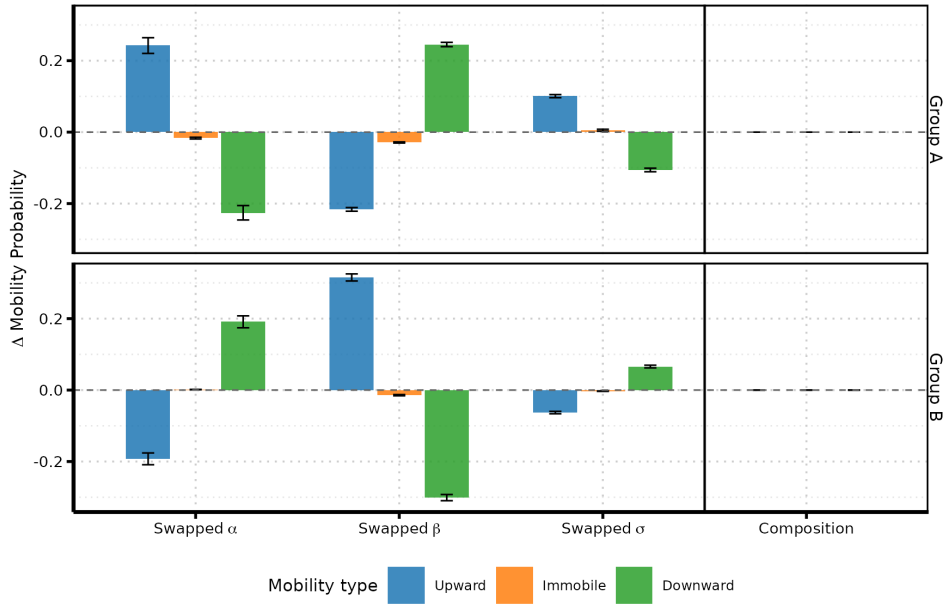

Figure 3: Parental-income-weighted posterior mean differences in group-level mobility probabilities under each counterfactual scenario. The final panel labeled "Composition" replaces the parental income distribution across groups while holding structural parameters fixed.

Four patterns emerge. First, intercept differences have substantial effects in Group A, increasing upward mobility and decreasing downward mobility by over 20 percentage points. The reverse effect in Group B is smaller, due to already-saturated downward mobility. Second, slope differences play the dominant role in Group B: adopting the steeper β_A increases upward mobility and reduces downward mobility by nearly 30 percentage points, while the reverse swap in Group A generates smaller (20 percentage point) effects. Third, variance swaps yield moderate changes of around 10 percentage points, enhancing upward mobility and dampening downward mobility in Group A, with weaker reverse effects in Group B. Finally, composition-only swaps have no effect, as expected in a simulation with identical parental income distributions.

These results validate the framework's ability to recover true generative parameters, generate interpretable structural decompositions, and quantify the mechanisms behind group-level mobil-

ity differences. Posterior sampling enables full distributional uncertainty to be carried through the counterfactual calculations, allowing credible intervals to be computed for both observed and hypothetical mobility probabilities. In doing so, the approach provides a flexible, simulation-based alternative to purely analytical decomposition—capable of disentangling complex, non-linear interdependencies between income floors, intergenerational persistence, volatility, and the distribution of origins.

8 Empirical Application: Black–White Differences in Absolute Mobility

Black–White disparities in intergenerational mobility provide a canonical context for applying the framework developed here. A long-standing interdisciplinary literature has documented persistent racial gaps in both relative and absolute mobility in the United States. Early sociological research characterized Black mobility as a "perverse" form of openness: weaker origin–destination associations did not translate into better outcomes because lower socioeconomic starting points meant upward mobility often produced only modest gains, while downward mobility from relatively advantaged positions remained common (Blau and Duncan, 1967; Duncan, 1968; Hout, 1984). More recent work has reaffirmed the persistence of racial mobility gaps and identified mechanisms including inequalities in income, wealth, education, family structure, labor markets, and residential segregation (Mazumder, 2012; Chetty et al., 2020; Bloome and Western, 2011; Bloome, 2014, 2017; Bloome and Opacic, 2024; Manduca, 2018). However, this literature typically examines mechanisms in isolation and varies in how mobility is conceptualized. Many studies equate absolute mobility with upward movement without accounting for downward or immobile trajectories (e.g., Chetty et al., 2020; Bloome and Opacic, 2024). The influential findings of Chetty et al. (2020)—showing that Black children experience lower upward and higher

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downward mobility than White children conditional on parental income—use national income rankings that make it harder to separate compositional differences in parental income from structural differences in the mobility process itself. A unified framework is needed to jointly model structural and compositional components and decompose their contributions. The empirical application below does precisely that, modeling absolute mobility probabilities as functions of group-specific parameters to clarify how income floors, intergenerational persistence, income uncertainty, and parental income jointly shape Black–White gaps in economic mobility.

For this application, I draw on data from the Panel Study of Income Dynamics (PSID), 1968–2019. The PSID began in 1968 with a nationally representative sample of approximately 18,000 individuals in 5,000 U.S. families and has followed these individuals and their descendants annually through 1997 and biennially thereafter. Its long panel structure, detailed intergenerational linkages, and rich labor-market information make it uniquely well suited for the study of intergenerational mobility.

Income is defined as labor earnings, including wages and salaries, the labor components of self-employment and farm income, and supplemental earnings such as bonuses, tips, and commissions. All income measures are adjusted for inflation. The analytic window includes observations between ages 25 and 60; for younger cohorts with shorter observation windows, income trajectories rely more heavily on early-career earnings. Children's income is measured as the natural logarithm of labor income at observed individual–time points. Permanent income is constructed as the individual-specific average of log labor income across years for individuals with at least two non-missing observations, explicitly accounting for within-individual uncertainty arising from limited observation length (described below). Permanent parental income is constructed using the full panel of inflation-adjusted labor income for both mothers and fathers. For each parent, I compute a centered seven-year rolling average, requiring at least five non-missing observations, and define permanent mid-career income as the value at age 42. A dominance criterion is applied

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such that parental income is defined as the maximum income at age 42 across both parents; for single-parent households, the measure is based on the sole parent. The analysis uses the natural logarithm of this permanent mid-career income. Reported zero labor income values are treated as missing throughout the income construction process.

The analytic sample consists of 12,445 Black and White sons and daughters drawn from original PSID families and their descendants. I restrict the sample to individuals born between 1952 and 1989 who appear as household heads or spouses in adulthood. The lower birth-year bound limits overrepresentation of early nest-leavers, while the upper bound ensures that even the youngest cohorts contribute at least five income observations at age 25 or older. Using these data, I implement an extended race-specific intergenerational income elasticity (IGE) framework in which both the conditional mean and the conditional variance of children's permanent income depend on parental income. This structure allows expected income levels and conditional income uncertainty to vary systematically across race and along the parental income distribution. Formally, let $g \in \{W, B\}$ index racial groups (White and Black). For each individual i in group g, denote parental permanent log-income by x_{ig} and the child's permanent log-income by y_{ig}. To facilitate interpretation of intercepts, parental log income is de-floored by subtracting the first percentile of the gender-specific parental income distribution, so that $x_{ig} = 0$ corresponds to this reference point.

Within this framework, the conditional distribution of child income is modeled as:

$$y_i \mid g, x_i \sim \text{Normal}\left(\mu_g(x_i),\ \sqrt{\eta_i^2/T_i + [\sigma_g(x_i)]^2}\right), \tag{35}$$

where the conditional mean is given by $\mu_{i[g]} = \alpha_g + \beta_g x_i$, with α_g capturing race-specific income floors and β_g measuring intergenerational persistence.

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The variance of observed permanent income is composed of two distinct elements. The first, η_i^2/T_i, captures individual-specific measurement uncertainty arising from limited income histories. Here, T_i denotes the number of non-missing income observations used to construct individual i's permanent income, and η_i^2 represents the within-individual variance of transitory log income. Individuals with shorter income histories therefore contribute noisier estimates of permanent income. The second component, $[\sigma_g(x_i)]^2$, represents the variance of children's permanent income conditional on parental income and group membership. This term captures genuine heterogeneity in long-run economic outcomes among individuals who share similar parental backgrounds, net of measurement noise. I model this conditional dispersion as $\sigma_{i[g]} \mid x_i \sim \text{LogNormal}\big(\gamma_g + \lambda_g x_i,\ \sigma_{\text{disp}}\big)$, where γ_g indexes baseline dispersion in children's permanent income within racial group g, λ_g governs how this dispersion varies with parental income, and σ_{disp} captures residual heterogeneity in dispersion unexplained by parental income or group membership.

Table 2: Posterior means and 95% credible intervals by gender.

		Women		Men	
	Parameter	Mean	95% CI	Mean	95% CI
Panel μ. Conditional mean parameters					
Black	α_B	9.72	(9.58, 9.86)	10.02	(9.88, 10.15)
	β_B	0.26	(0.19, 0.33)	0.25	(0.19, 0.32)
White	α_W	9.72	(9.60, 9.85)	10.31	(10.21, 10.42)
	β_W	0.32	(0.27, 0.37)	0.25	(0.21, 0.29)
Panel σ. Conditional dispersion parameters					
Black	γ_B	-0.50	(-0.66, -0.33)	-0.57	(-0.74, -0.36)
	λ_B	-0.05	(-0.14, 0.03)	-0.07	(-0.17, 0.01)
White	γ_W	-0.45	(-0.61, -0.29)	-0.61	(-0.77, -0.46)
	λ_W	-0.04	(-0.10, 0.02)	-0.04	(-0.11, 0.02)
LOOIC		47,813.4		46,839.7	
WAIC		47,756.3		46,768.9	

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Table 2 reports posterior means and 95% credible intervals for all model parameters, revealing substantial heterogeneity in intergenerational income transmission across both race and gender. Among women, Black and White daughters share an identical income floor ($\alpha_B = \alpha_W = 9.72$), indicating that daughters born to parents at the very bottom of the income distribution are expected to attain similar income levels in adulthood regardless of race. Racial differences emerge, however, in the strength of intergenerational persistence: parental income is more strongly associated with adult earnings among White than Black daughters ($\beta_W = 0.32$ vs. $\beta_B = 0.26$). As a consequence, income advantages at higher parental ranks translate more fully into adult earnings for White women, and racial inequality among daughters widens as one moves up the parental income distribution. Despite this stronger dependence of expected income on parental income, White daughters exhibit slightly higher baseline income unpredictability than Black daughters ($\gamma_W = -0.45$ vs. $\gamma_B = -0.50$), although conditional income uncertainty declines at similar rates with parental income for both groups ($\lambda_W = -0.04$ vs. $\lambda_B = -0.05$).

Among men, intergenerational income transmission follows a different pattern. White sons begin adulthood from a substantially higher income floor than Black sons ($\alpha_W = 10.31$ vs. $\alpha_B = 10.02$), while the strength of intergenerational persistence is virtually identical across race ($\beta_W = \beta_B = 0.25$). This configuration implies that racial gaps among men are anchored at the bottom of the income distribution and then reproduced in parallel as parental income rises. At the same time, Black men face greater conditional income uncertainty: baseline income unpredictability is higher for Blacks than for Whites ($\gamma_B = -0.57$ vs. $\gamma_W = -0.61$), even though uncertainty declines more steeply with parental income among Blacks ($\lambda_B = -0.07$ vs. $\lambda_W = -0.04$). Consequently, Black sons experience both lower expected incomes and more uncertain income transmission across much of the parental income distribution.

Substantively, these estimates delineate two distinct mobility regimes that are partially consistent with Hout (1984)'s notion of perverse openness. Among Black Americans, equal or lower income

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floors combined with equal or weaker intergenerational persistence—and, among men, elevated baseline conditional income uncertainty—produce a mobility regime that is formally open but substantively constrained. By contrast, the mobility regime observed among Whites—particularly White men—is less exposed to downward risk conditional on origins. Equal or higher income floors, equal or stronger intergenerational persistence, and lower baseline conditional income uncertainty imply that the transmission process yields a tighter and more predictable reproduction of advantage, in which higher parental income tends to translate into higher and more secure adult outcomes across generations.

Building on these estimates, I compute factual conditional absolute mobility distributions and conduct a counterfactual parameter-swap simulation to identify the structural sources of Black–White differences in absolute mobility. Using posterior draws of $\alpha_g, \beta_g, \gamma_g, \lambda_{g_{g \in W,B}}$, I generate counterfactual mobility distributions by simulating child incomes under alternative parameter assignments. Figure 4 summarizes the results, showing observed mobility patterns by group and gender (Panels A–B) and the corresponding changes in mobility probabilities induced by each parameter swap (Panels C–D).

Among women, absolute mobility patterns differ by race primarily at higher parental income levels. At the bottom of the parental income distribution, Black and White women display similar probabilities of upward mobility, making upward movement the modal outcome for both groups. Because Black women are disproportionately concentrated in disadvantaged origins, upward mobility from the bottom is more frequently experienced among Black women in the population as a whole. At middle parental income levels, downward mobility becomes more likely for women of both races, with only modest racial differences in the distribution across mobility states. Racial divergence is most pronounced at the top of the parental income distribution. There, downward mobility overwhelmingly dominates for both groups, but it is more prevalent among Black women. Lower returns to affluent origins—reflecting weaker intergenerational

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persistence—make matching or exceeding parental income especially unlikely. As a result, Black women from high-income families face a higher risk of downward mobility than their White counterparts.

Among men, racial differences in absolute mobility are driven by systematically higher expected income levels among White men across the parental income distribution. At low parental income levels, upward mobility is the modal outcome for both Black and White men, but it is more frequently realized among Black men because they are more heavily represented at the bottom of the parental income distribution. At middle parental income levels, Black and White men exhibit broadly similar probabilities of upward, immobile, and downward mobility; however, White men are more likely to either move upward or remain immobile, while Black men face a higher risk of downward movement. Racial differences become more pronounced at higher parental income levels. There, White men are less likely to experience downward mobility and more likely to maintain or modestly exceed parental income levels. Consequently, Black men from affluent origins face a higher risk of downward mobility, whereas White men are relatively insulated from the loss of parental advantage. Because White men are more likely to originate from middle- and high-income families, these mobility patterns are especially consequential for their overall mobility profile.

The counterfactual scenarios in Panel C and D of Figure 4 further clarify the contribution of each structural component to Black–White differences in absolute mobility, and how these contributions vary by gender. Differences in relative immobility emerge as the dominant source of divergence in absolute mobility among women, while differences in income floors drive racial absolute mobility gaps among men.

For women, racial differences in intergenerational persistence account for most of the Black–White gap in absolute mobility. Counterfactual slope swaps indicate that if Black women were to exhibit the stronger income transmission observed among White women, their probability of upward

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mobility would increase, with corresponding reductions in downward mobility. These effects are concentrated among women from middle-income parental origins, where upward mobility rises and downward mobility falls most sharply. While the direction of these changes is clear, wide credible intervals limit the precision with which their magnitude can be quantified. At the bottom of the parental income distribution, slope swaps have little effect on absolute mobility, as expected incomes already lie well above the upward mobility threshold, leaving limited scope for additional gains. At the top of the distribution, effects again attenuate, since downward mobility is already the dominant outcome for both groups. The reverse counterfactual—assigning Black women's lower persistence to White women—produces a mirror-image pattern, reducing upward mobility and increasing downward mobility most strongly at middle parental income levels.

For men, Black–White differences in absolute mobility are driven entirely by differences in income floors. Counterfactual income-floor swaps show that if Black men were to exhibit the higher income floor observed among White men, their probability of upward mobility would increase, with a corresponding reduction in downward mobility. These effects are most pronounced among men from middle-income parental backgrounds, where upward mobility rises sharply and downward mobility falls by similar magnitudes. At higher parental income levels, income-floor swaps mainly increase the likelihood that Black men remain at the top of the distribution, rather than generating additional upward mobility. By contrast, at the bottom of the parental income distribution, income-floor swaps have little effect on absolute mobility, as expected incomes already lie well above the upward-mobility threshold. Effects also attenuate at the top of the distribution, where upward mobility thresholds lie far above expected incomes and downward mobility is already the dominant outcome. The reverse counterfactual yields a mirror-image pattern for White men, increasing downward mobility and reducing the likelihood of maintaining or exceeding parental income levels.

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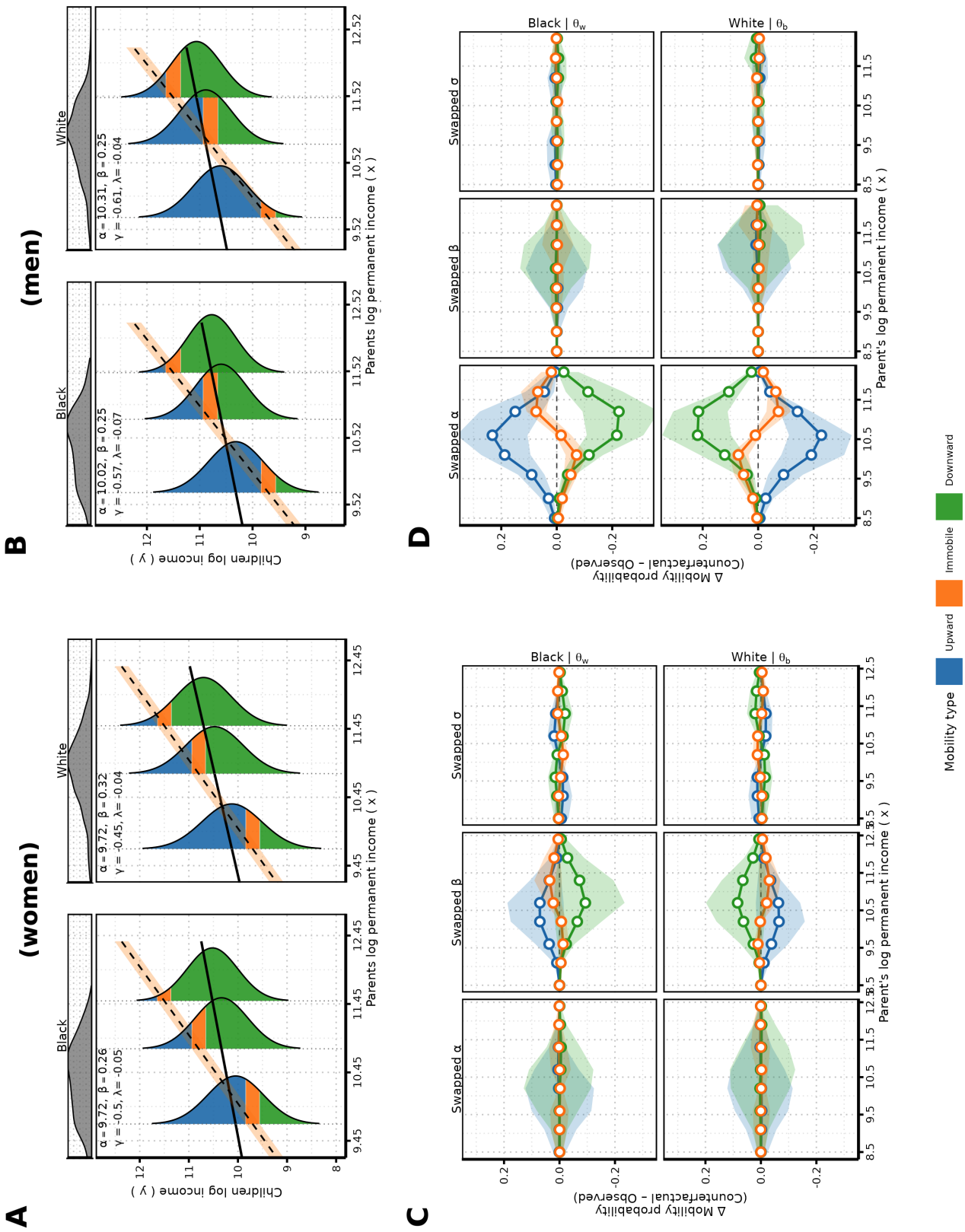

Figure 4: Panels **A** and **B** display posterior mean probabilities of upward, downward, and immobile mobility across parental income, separately by group, for women and men, respectively. Panels **C** and **D** report pointwise counterfactual effects—defined as differences between counterfactual and observed mobility probabilities—along with 95% credible intervals, by group and parameter swap, again for women and men.

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The results above report pointwise factual and counterfactual absolute mobility probabilities at each level of parental income x, illustrating how each structural parameter shapes mobility conditional on family origin. Conditional patterns, however, do not translate mechanically into group-level mobility differences. This is especially consequential in the Black–White context, where parental income distributions are highly unequal and Black families are disproportionately concentrated at the lower end of the income distribution. Because the effects of structural parameters vary sharply with parental income, their impact on population-level absolute mobility also depends on how individuals in each group are distributed along the parental income distribution.

To assess these group-level consequences, I aggregate the pointwise counterfactual effects by weighting income-specific mobility differences by the empirical share of families at each parental income level. Using each group's observed parental income distribution, I average the predicted mobility changes attributable to each structural component (α, β, σ) across the income ladder. This procedure maps conditional effects into average group-level effects on mobility. Because parameter effects are highly origin-dependent, the aggregate impact of each parameter reflects both its local influence and the concentration of families along the income distribution. Finally, to quantify the role of origin composition alone, I simulate counterfactual aggregate mobility holding each group's structural parameters fixed while swapping their parental income distributions. This exercise asks how absolute mobility would change if Black and White families retained their observed mobility mechanisms (α, β, σ) but were placed in each other's parental income distributions.

Figure 5 reports the resulting parental-income-weighted posterior mean differences (bars) and their associated uncertainty intervals (whiskers) across the three parameter-swap scenarios, along with the composition-only counterfactual:

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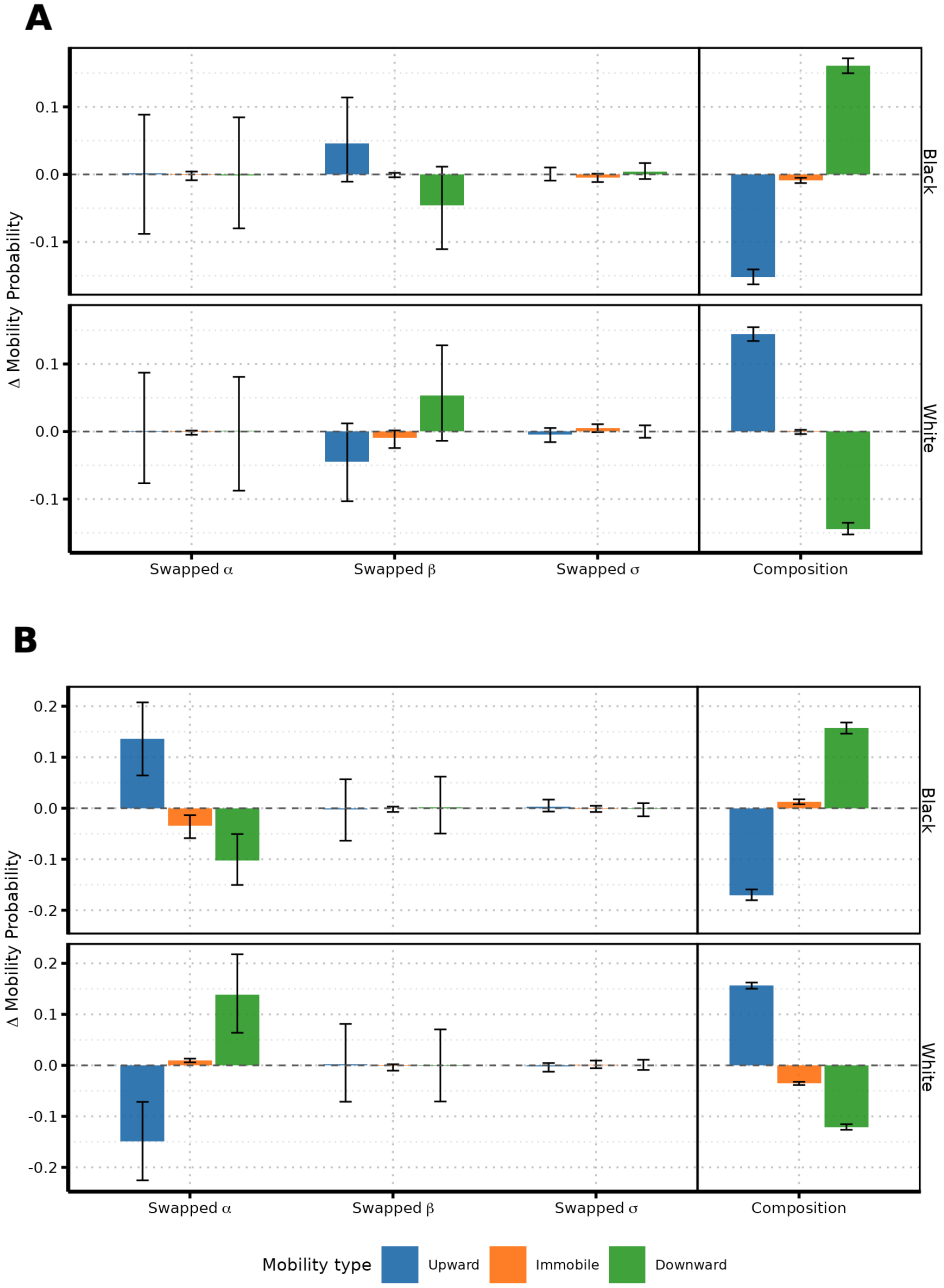

Figure 5: Parental-income-weighted posterior mean differences in group-level mobility probabilities under each counterfactual scenario. The final panel labeled "Composition" replaces the parental income distribution across groups while holding structural parameters fixed.

For women (Panel A), differences in intergenerational persistence generate only modest average effects on absolute mobility. Steepening Black women's intergenerational slope to match that

of White men increases upward mobility by roughly 5 percentage points on average and reduces downward mobility by a similar amount. Flattening White women's slope to the level observed among Black women produces a mirror-image pattern of comparable magnitude. However, these average effects are not statistically significant. For men (Panel B), income-floor differences produce substantially larger changes in absolute mobility. If Black men were to start from the higher income floor observed among White men, their probability of upward mobility would increase by more than 10 percentage points on average, with a corresponding reduction in downward mobility. Conversely, if White men faced the lower income floor observed among Black men, their probability of downward mobility would rise by roughly 15 percentage points, while upward mobility would decline by a similar amount.

The most consequential results concern composition, echoing findings by Bloome and Opacic (2024). Swapping only the parental income distributions—while holding all structural parameters fixed—produces large and systematic shifts in aggregate mobility across race and gender. If Black children were drawn from the more affluent parental income distribution of White families, both Black sons and daughters would experience a substantial decline in upward mobility and a corresponding increase in downward mobility, on the order of 12–15 percentage points. The reverse counterfactual generates a mirror-image pattern for White children: drawing them from the comparatively poorer parental income distribution of Black families would increase their probability of upward mobility and reduce downward mobility by similar magnitudes. The substantive interpretation is straightforward but consequential. Part of the higher observed rate of upward mobility among Black Americans reflects the arithmetic fact that surpassing one's parents is mechanically easier when parental incomes are low. Conversely, part of the higher incidence of downward mobility among White Americans reflects the difficulty of matching or exceeding very high parental incomes (Bloome and Opacic, 2024). Holding everything else constant, disadvantaged origins mechanically facilitate upward mobility, while affluent origins mechanically

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expand the scope for downward mobility. Failing to separate composition from structure risks interpreting mechanically induced upward mobility as evidence of opportunity.

Taken together, these results show that similar observed rates of absolute mobility can conceal fundamentally different underlying mechanisms. For men, racial differences in expected income—driven by disparities in income floors combined with nearly identical levels of intergenerational persistence—shape mobility prospects in one direction, while differences in the composition of parental origins operate in the opposite direction, partially offsetting these structural effects in the aggregate. For women, such offsetting dynamics are far weaker: income floors are similar across race, and differences in intergenerational transmission are modest, leaving limited scope for structural forces to counterbalance compositional ones. Importantly, the relatively simple balance of forces observed here is specific to this empirical setting. In other contexts, multiple structural parameters and compositional differences may interact in more complex ways, producing aggregate mobility patterns that are difficult to infer from parameter values alone—a point illustrated more generally by the Monte Carlo experiments in Section 7.

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9 Conclusion

This paper develops a unified framework for intergenerational mobility by embedding absolute and relative mobility within the same income-generating process. Rather than treating these dimensions as separate or competing summaries of stratification, the analysis shows that both arise from the joint configuration of income levels, intergenerational persistence, transmission uncertainty, and the distribution of parental resources. This integration clarifies why absolute and relative mobility can diverge in systematic ways and why neither can be meaningfully interpreted in isolation from the other.

A central implication is that relative mobility—captured by the strength of the origin–destination

association—does not translate mechanically into favorable or unfavorable absolute mobility outcomes. Its consequences depend critically on where expected outcomes lie relative to parental benchmarks and on the degree of uncertainty surrounding income transmission. As a result, the same level of intergenerational persistence can generate very different absolute mobility regimes across populations and across segments of the parental income distribution. This helps explain why societies, cohorts, or groups with similar levels of relative mobility may nonetheless experience sharply different patterns of upward and downward movement, and why changes in relative mobility alone are often insufficient to account for observed shifts in absolute mobility.

The framework also provides a principled basis for decomposing group differences in absolute mobility. By grounding counterfactual comparisons in an explicit generative model, the parameter-swap decomposition isolates the contributions of income floors, intergenerational persistence, and transmission uncertainty to observed mobility gaps, while separately accounting for compositional differences in parental income distributions.

The empirical application to Black–White mobility in the United States illustrates the substantive payoff of this integration. The results show that similar or weaker intergenerational persistence among Black Americans does not necessarily imply more favorable absolute mobility prospects. Relative to Whites, comparable or weaker persistence combined with similar or lower income floors yields lower expected incomes for Black children across most parental income origins. At the same time, comparatively poorer parental origins mechanically increase the likelihood that Black individuals surpass their parents' income, while fewer Black individuals originate from high-income families, where the scope for downward mobility is greatest. The resulting mobility pattern is one in which upward mobility from low-income origins is common among Black Americans, while downward mobility from middle- and high-income origins is also more frequent than among Whites. This configuration aligns with what has been described as *perverse openness*: apparent fluidity in relative terms paired with persistent economic vulnerability (Blau

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and Duncan, 1967; Duncan, 1968; Hout, 1984).

More broadly, these findings underscore the limits of one-dimensional accounts of intergenerational mobility. Absolute mobility cannot be understood solely as a consequence of growth or inequality, nor can relative mobility be interpreted independently of income levels and risk. Policies or institutional changes that affect persistence, income floors, or uncertainty may have qualitatively different implications for mobility depending on where families are located in the parental income distribution. Evaluating mobility regimes therefore requires attention not only to average rates of upward movement, but also to the mechanisms that produce them and to the distribution of risks they entail.

The framework developed here is intentionally parsimonious but also readily extensible. It can be applied to other forms of mobility, such as education or wealth, to different group comparisons, and to cross-national contexts where growth, inequality, and uncertainty vary sharply. Methodologically, the approach can be extended to richer functional forms or to semi-parametric and nonparametric settings. More generally, modeling mobility as a probabilistic process brings absolute and relative mobility into a common analytic space and allows them to be interpreted as complementary—rather than competing—dimensions of intergenerational inequality.

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Code, Materials, and Data Availability: All data and code necessary to reproduce the analyses, tables, and figures reported in this article are publicly available [here]. The repository contains fully scripted workflows enabling complete replication of results.

Preregistration Statement: This study was not preregistered.

AI Use Statement: Generative AI tools were used for language editing, proofreading, and coding assistance. All theoretical development, empirical analyses, and substantive interpretations are solely the author's responsibility.

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References

Berman, Y. (2022). The Long-Run Evolution of Absolute Intergenerational Mobility. *American Economic Journal: Applied Economics*, 14(3):61–83.

Black, S. and Devereux, P. (2010). Recent Developments in Intergenerational Mobility. *National Bureau of Economic Research*.

Blau, P. M. and Duncan, O. D. (1967). *American Occupational Structure*. Wiley, New York,. Publication Title: The American Occupational Structure ISSN: 00031224.

Bloome, D. (2014). Racial Inequality Trends and the Intergenerational Persistence of Income and Family Structure. *American Sociological Review*, 79(6):1196–1225. Publisher: NIH Public Access ISBN: 0003-1224.

Bloome, D. (2017). Childhood Family Structure and Intergenerational Income Mobility in the United States. *Demography*, 54(2):541–569.

Bloome, D. (2025). Absolute and Relative Mobility: Two Frameworks for Connecting Intergenerational Mobility in Absolute and Relative Terms. *Sociological Methods & Research*, 54(4):1252–1288.

Bloome, D. and Opacic, A. (2024). Absolute income mobility obscures marginalized children's disadvantages. *Proceedings of the National Academy of Sciences*, 121(25):e2321418121.

Bloome, D. and Western, B. (2011). Cohort change and racial differences in educational and income mobility. *Social Forces*, 90(2):375–395. Publisher: Oxford University Press.

Boudon, R. (1973). *Mathematical structures of social mobility.* Jossey-Bass, San Francisco, [1st ed.] edition. Open Library ID: OL5430493M.

Boudon, R. (1974). *Education, Opportunity, and Social Inequality: Changing Prospects in Western Society. -*. John Wiley & Sons Canada, Limited. Series Title: Urban Research Series.

Breen, R. (2010). Social Mobility and Equality of Opportunity Geary Lecture Spring 2010. *The Economic and Social Review*, 41:413–428.

Chetty, R., Grusky, D., Hell, M., Hendren, N., Manduca, R., and Narang, J. (2017). The fading American dream: Trends in absolute income mobility since 1940. *Science*, 356(April):398–406.

Chetty, R., Hendren, N., Jones, M. R., and Porter, S. R. (2020). Race and Economic Opportunity in the United States: an Intergenerational Perspective*. *The Quarterly Journal of Economics*, 135(2):711–783.

Cobalti, A. (1988). Alternative conceptual frameworks for the analysis of mobility tables and the log-linear models. *Quality and Quantity*, 22(1):31–47.

Corak, M. (2013). Income Inequality, Equality of Opportunity, and Intergenerational Mobility. *The Journal of Economic Perspectives*, pages 79–102. Publisher: JSTOR.

Duncan, O. D. (1968). Patterns of Occupational Mobility Among Negro Men. *Demography*, 5(1):11–22.

Fields, G. S. (2010). "But That Is Not What Economic Mobility Is! ".

For Peer Review

Fields, G. S. and Ok, E. A. (1996). The Meaning and Measurement of Income Mobility. *Journal of Economic Theory*, 71(2):349–377.

Goldthorpe, J. H. (2016). Social class mobility in modern Britain: changing structure, constant process. *Journal of the British Academy*, 4(89-111).

Hout, M. (1983). *Mobility Tables*. Sage Research Methods.

Hout, M. (1984). Occupational Mobility of Black Men: 1962 to 1973. *American Sociological Review*, 49(3):308. ISBN: 00031224.

Manduca, R. (2018). Income inequality and the persistence of racial economic disparities. *Sociological Science*, 5:182–205.

Mazumder, B. (2012). Black-White Differences in Inter-Generational Economic Mobility in the U.S. Technical report. Publication Title: SSRN Electronic Journal Volume: XXXVIII Issue: 1.

Mitnik, P. A., Bryant, V., and Weber, M. (2019). The intergenerational transmission of family-income advantages in the United States. *Sociological Science*, 6:380–415. Publisher: Society for Sociological Science.

Sobel, M. E. (1983). Structural Mobility, Circulation Mobility and the Analysis of Occupational Mobility: A Conceptual Mismatch. *American Sociological Review*, 48(5):721–727.

Sobel, M. E., Hout, M., and Duncan, O. D. (1985). Exchange, Structure, and Symmetry in Occupational Mobility. *American Journal of Sociology*, 91(2):359–372.

Solon, G. (1992). Intergenerational Income Mobility in the United States. *The American Economic Review*, 82(3):393–408.

For Peer Review

Solon, G. (1999). Intergenerational Mobility in the Labor Market. Volume 3,:1761–1800. ISBN: 1573-4463.

Xie, Y. (1992). The Log-Multiplicative Layer Effect Model for Comparing Mobility Tables. *American Sociological Review*, 57(3):380. Publisher: JSTOR ISBN: 0003-1224.

Zimmerman, D. J. (1992). Regression toward mediocrity in economic stature. *The American Economic Review*, pages 409–429. Publisher: JSTOR.

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10 Appendix A. Derivation of Marginal Effects on Absolute Mobility Probabilities

This appendix derives the marginal effects of key structural parameters—intercept (α), slope (β), dispersion (σ), and parental income (x)—on the absolute mobility probabilities defined in the main text. These effects clarify how changes in the income-generating process influence the likelihood that a child experiences upward, downward, or immobile economic trajectories.

I begin by recalling the definitions:

$$\Pr\big(M^{\uparrow}\big) = \Pr(y > x + \delta \mid x, g) = 1 - F(z^{\uparrow}), \quad \text{where} \quad z^{\uparrow} = \frac{x + \delta - \mu}{\sigma}$$

$$\Pr\big(M^{\downarrow}\big) = \Pr(y < x - \delta \mid x, g) = F(z^{\downarrow}), \quad \text{where} \quad z^{\downarrow} = \frac{x - \delta - \mu}{\sigma}$$

$$\Pr\big(M^{0}\big) = F(z^{\uparrow}) - F(z^{\downarrow})$$

with $\mu = \alpha + \beta x$, and $F(\cdot)$ denoting the cumulative distribution function of the outcome distribution, and $f(\cdot)$ its associated probability density function.

Effect of the intercept α.

Upward mobility depends on α through the mean $\mu = \alpha + \beta x$. Applying the chain rule:

$$\frac{\partial \Pr\big(M^{\uparrow}\big)}{\partial \alpha} = \frac{\partial}{\partial \alpha}\left[1 - F(z^{\uparrow})\right] = -f(z^{\uparrow}) \cdot \frac{\partial z^{\uparrow}}{\partial \alpha}.$$

This step follows from the fact that the derivative of any cumulative distribution function $F(z)$

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is the corresponding density function $f(z)$, so differentiating $1 - F(z^{\uparrow})$ with respect to $z^{\uparrow}$ yields $-f(z^{\uparrow})$.

Next, since $z^{\uparrow} = \frac{x+\delta-(\alpha+\beta x)}{\sigma}$, I find:

$$\frac{\partial z^{\uparrow}}{\partial \alpha} = -\frac{1}{\sigma},$$

and therefore:

$$\frac{\partial \Pr\left(M^{\uparrow}\right)}{\partial \alpha} = \frac{f(z^{\uparrow})}{\sigma}.$$

By similar reasoning:

$$\frac{\partial \Pr\left(M^{\downarrow}\right)}{\partial \alpha} = -\frac{f(z^{\downarrow})}{\sigma}, \quad \frac{\partial \Pr\left(M^{0}\right)}{\partial \alpha} = \frac{f(z^{\uparrow}) - f(z^{\downarrow})}{\sigma}.$$

Effect of the slope β.

Again applying the chain rule:

$$\frac{\partial \Pr\left(M^{\uparrow}\right)}{\partial \beta} = -f(z^{\uparrow}) \cdot \frac{\partial z^{\uparrow}}{\partial \beta}.$$

Since $\mu = \alpha + \beta x$, I obtain:

$$\frac{\partial z^{\uparrow}}{\partial \beta} = -\frac{x}{\sigma}, \quad \text{so} \quad \frac{\partial \Pr\left(M^{\uparrow}\right)}{\partial \beta} = \frac{x f(z^{\uparrow})}{\sigma}.$$

Likewise:

$$\frac{\partial \Pr\left(M^{\downarrow}\right)}{\partial \beta} = -\frac{x f(z^{\downarrow})}{\sigma}, \quad \frac{\partial \Pr\left(M^{0}\right)}{\partial \beta} = \frac{x\left[f(z^{\downarrow}) - f(z^{\uparrow})\right]}{\sigma}.$$

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Effect of the dispersion σ.

For dispersion, both $z^{\uparrow}$ and $z^{\downarrow}$ depend explicitly on σ, so I again apply the chain rule:

$$\frac{\partial z^{\uparrow}}{\partial \sigma} = -\frac{x+\delta-\mu}{\sigma^2} = -\frac{z^{\uparrow}}{\sigma},$$

and thus:

$$\frac{\partial \Pr\left(M^{\uparrow}\right)}{\partial \sigma} = -f(z^{\uparrow}) \cdot \left(-\frac{z^{\uparrow}}{\sigma}\right) = \frac{z^{\uparrow} f(z^{\uparrow})}{\sigma}.$$

Similarly:

$$\frac{\partial \Pr\left(M^{\downarrow}\right)}{\partial \sigma} = -\frac{z^{\downarrow} f(z^{\downarrow})}{\sigma}, \quad \frac{\partial \Pr\left(M^{0}\right)}{\partial \sigma} = \frac{-z^{\uparrow} f(z^{\uparrow}) + z^{\downarrow} f(z^{\downarrow})}{\sigma}.$$

Effect of parental income x.

To compute $\frac{\partial \Pr\left(M^{\uparrow}\right)}{\partial x}$, note that $z^{\uparrow} = \frac{x+\delta-(\alpha+\beta x)}{\sigma} = \frac{(1-\beta)x+\delta-\alpha}{\sigma}$, so

$$\frac{\partial z^{\uparrow}}{\partial x} = \frac{1-\beta}{\sigma},$$

and then:

$$\frac{\partial \Pr\left(M^{\uparrow}\right)}{\partial x} = -f(z^{\uparrow}) \cdot \frac{\partial z^{\uparrow}}{\partial x} = \frac{\beta-1}{\sigma} f(z^{\uparrow}).$$

Analogously:

$$\frac{\partial \Pr\left(M^{\downarrow}\right)}{\partial x} = \frac{1-\beta}{\sigma} f(z^{\downarrow}), \quad \frac{\partial \Pr\left(M^{0}\right)}{\partial x} = \frac{1-\beta}{\sigma}[f(z^{\uparrow}) - f(z^{\downarrow})].$$

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11 Appendix B. Hierarchical Model Specification and Prior Details

The model specifies the conditional distribution of children's income given parental income, allowing both expected outcomes and income uncertainty to vary by group and along the parental income distribution. Let y_i and x_i denote child and parental permanent income on the level scale, respectively, and let $g \in \{1, 2\}$ index group membership. Income is modeled using a lognormal likelihood, with the conditional mean specified on the log scale:

```
for (i in 1:N) {
  y[i] ~ lognormal(mu[i], sigma_total[i]);
}
```

Here, `mu[i]` denotes the conditional mean of log income and `sigma_total[i]` the total conditional standard deviation entering the likelihood.

Conditional mean. The conditional mean is specified as a group-specific linear function of centered parental log income:

```
for (i in 1:N) {
  int g = group[i] + 1;
  real logx_ctr = log(x[i]) - x_ref_log;
  mu[i] = alpha[g] + beta[g] * logx_ctr;
}
```


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The parameter `alpha[g]` captures a group-specific income floor, while `beta[g]` is the intergenerational elasticity governing the strength of income transmission. Parental income is centered at `x_ref_log`, corresponding to a fixed reference point on the log scale, which facilitates interpretation of intercepts.

Conditional dispersion. Income uncertainty is modeled through a heteroskedastic log-standard-deviation function:

```
sigma[i] = exp(gamma[g] + lambda[g] * logx_ctr
               + sigma_dispersion * sigma_raw[i]);
```

The parameter `gamma[g]` governs baseline dispersion, `lambda[g]` allows dispersion to vary with parental income, and `sigma_dispersion` scales idiosyncratic volatility through `sigma_raw[i]` $\sim$ Normal$(0, 1)$. This specification allows for modest individual-level heterogeneity in dispersion while preserving numerical stability.

Measurement error and total variance. Observed permanent income is constructed from a finite number of income realizations. Accordingly, the standard deviation entering the likelihood combines structural dispersion and measurement uncertainty:

```
for (i in 1:N) {
  sigma_total[i] = sqrt(square(sigma[i])
                        + square(logy_sd[i] / sqrt(T_obs[i])));
}
```

The term `logy_sd[i] / sqrt(T_obs[i])` captures residual noise arising from finite income histories, with uncertainty declining as the number of observed periods increases. In the limiting

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case where permanent income is observed without error ($T_i \to \infty$), the measurement component vanishes and the conditional standard deviation simplifies to $\sigma_g(x)$, which represents the structural dispersion of permanent income conditional on parental income and group membership.

Hierarchical structure. Group-specific parameters are modeled hierarchically using a non-centered parameterization:

```
vector[2] alpha  = alpha_mean  + alpha_sd  * alpha_raw;
vector[2] beta   = beta_mean   + beta_sd   * beta_raw;
vector[2] gamma  = gamma_mean  + gamma_sd  * gamma_raw;
vector[2] lambda = lambda_mean + lambda_sd * lambda_raw;
```

Raw deviations follow standard normal priors:

```
alpha_raw  ~ normal(0, 1);
beta_raw   ~ normal(0, 1);
gamma_raw  ~ normal(0, 1);
lambda_raw ~ normal(0, 1);
```

This formulation pools information across groups while permitting group-level deviations, improving posterior conditioning and sampling efficiency.

Priors. Population-level means are assigned weakly informative but empirically grounded priors:

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```
alpha_mean  ~ normal(0, 0.7);
beta_mean   ~ normal(0.5, 0.2);
gamma_mean  ~ normal(0, 0.7);
lambda_mean ~ normal(0, 0.2);
```

The prior on `beta_mean` reflects evidence from U.S. studies, where intergenerational elasticities typically lie between 0.3 and 0.6, with most estimates around 0.4–0.5. Priors on dispersion parameters encode expectations of moderate baseline inequality and weak heteroskedasticity.

Population-level standard deviations governing between-group heterogeneity are assigned half-normal priors:

```
alpha_sd  ~ normal(0, 0.5);   // half-normal
beta_sd   ~ normal(0, 0.2);
gamma_sd  ~ normal(0, 0.3);
lambda_sd ~ normal(0, 0.1);
```

where positivity is enforced through parameter constraints. The auxiliary dispersion parameters are specified as:

```
sigma_dispersion ~ normal(0, 0.15);
sigma_raw ~ normal(0, 1);
```

introducing mild, regularized heterogeneity in conditional volatility.

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Likelihood and predictive quantities. The likelihood for individual i is given by:

```
y[i] ~ lognormal(mu[i], sigma_total[i]);
```

Modeling income on the level scale with a lognormal likelihood preserves interpretability in terms of proportional changes while remaining consistent with standard practice in intergenerational mobility research.

The `generated quantities` block produces posterior predictive draws and diagnostic quantities:

```
for (i in 1:N) {
  y_pred[i] = lognormal_rng(mu[i], sigma_total[i]);
  residual[i] = log(y[i]) - mu[i];
  log_lik[i] = lognormal_lpdf(y[i] | mu[i], sigma_total[i]);
}
```

and computes counterfactual mobility probabilities under four parameter configurations: (i) observed group parameters; (ii) counterfactual income floors (`alpha` swapped across groups); (iii) counterfactual persistence (`beta` swapped); and (iv) counterfactual dispersion (`gamma` and `lambda` swapped).

For a given parental income level and tolerance parameter `epsilon`, mobility probabilities are computed as:

```
real up_bound  = log(x_new[j]) + epsilon;
real low_bound = log(x_new[j]) - epsilon;
```


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```

up_obs[j]   = 1 - normal_cdf(up_bound | mu_obs, sigma_obs);
down_obs[j] = normal_cdf(low_bound | mu_obs, sigma_obs);
imm_obs[j]  = normal_cdf(up_bound | mu_obs, sigma_obs)
              - normal_cdf(low_bound | mu_obs, sigma_obs);
```

Counterfactual scenarios are constructed by swapping parameters across groups while holding others fixed:

```
int g     = group_new[j] + 1;
int other = 3 - g;

real mu_alpha      = alpha[other] + beta[g] * logx_ctr_new;
real mu_beta       = alpha[g] + beta[other] * logx_ctr_new;
real sigma_shared  = exp(gamma[other] + lambda[other] * logx_ctr_new);
```

These posterior mobility probabilities constitute the inputs for the decomposition analyses reported in Section 5, attributing observed mobility differences to income floors (`alpha`), intergenerational persistence (`beta`), and transmission uncertainty (`gamma`, `lambda`).

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12 Appendix C. PSID Analytic Sample Details

Table 3: Descriptive Statistics by Racial Group and Gender

	Women		Men	
	Black ($g = 0$)	White ($g = 1$)	Black ($g = 0$)	White ($g = 1$)
Parental income (level)	35,355	64,929	37,459	63,871
Child income (level)	30,135	41,368	38,256	66,088
Upward mobility share	0.338	0.231	0.441	0.423
Immobile share	0.141	0.134	0.168	0.180
Downward mobility share	0.522	0.635	0.390	0.397

Notes: Parental and child incomes are measured on the level scale in constant dollars. Reported statistics are unconditional means and standard deviations computed within each racial group. No sampling weights are applied. The parental log-income floor is fixed at $x_0 = 9.617$ (corresponding to a level income of \$15,011), and the absolute mobility threshold is set to $\delta = \log(1.15)$; both are fixed by construction and therefore omitted from the table.

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